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The Mediating Effects of Social Capital in the Relationship between Entrepreneurial Orientation and Startup Performance

: The cases from the Philippines

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창업성과에 영향을 미치는 기업가적 지향성과 스타트업의 사회적 자본에 관한 매개효과 연구

: 필리핀 스타트업을 중심으로

The Mediating Effects of Social Capital in the Relationship between Entrepreneurial Orientation and Startup Performance

: The cases from the Philippines

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Abstract

Based on innovative technologies and new start-up ideas, startups have been actively researched on the entrepreneurship needed to strengthen the organization's capabilities and generate results in the start-up process. This study studied the entrepreneurial orientation on the performance of startups in the Philippines and the financial and non-financial performance of enterprises. This study carried out not only the entrepreneurial orientation to the performance of Filipino startups but also the role of social capital as parameters in the performance of enterprises. The empirical research was completed for 93 Philippine startups and the suitability of the research model was evaluated with a PLS-based structural equation model. The results of the study first confirmed that the enterprise orientation of Philippine startups has a positive impact on both financial and non-financial performance of the enterprises. Second, the entrepreneurial orientation of Philippine startups has been shown to have a positive effect on both the structural, cognitive and relational dimensions of social capital. Third, it was found that the relevant dimensions of social capital mediated both the corporate orientation and the relationship between the financial and non-financial performance of the entity. Entrepreneurial orientation has been confirmed to be directly or indirectly affecting the performance of startups through social capital. These findings reaffirmed that entrepreneurial orientation is still a valid important factor in developing countries as well as in countries such as Korea and the United States. Based on this study, we have identified the need for research from a more integrated perspective, such as the concept of strategic orientation. Finally, practical implications were presented to reflect the findings analyzed.

KeyWords: Philippines, Startup, Entrepreneurial Orientation, Social Capital, PLS-SEM

국문초록

스타트업은 혁신적 기술과 새로운 창업 아이디어를 바탕으로 창업하는 과정에서 조직의 역량 강화 및 성과를 창출해야 하는데, 그 과정에 필수불가결한 기업가정신에 대해 지금까지 활발한 연구가 수행되어 왔다. 본 연구는 필리핀 스타트업을 대상으로 스타트업의 성과에 미치는 기업가적 지향성과 기업의 재무적 및 비재무적 성과에 관하여 연구하였다. 이뿐만 아니라 사회적 자본을 매개변수로 하여 이것이 기업의 성과에서 어떠한 역할을 하는지에 대해 연구를 수행하였다. 이에 관하여 필리핀 스타트업 93 개의 기업을 대상으로 실증연구를 완료하였으며, PLS 기반의 구조방정식모델로 연구 모형의 적합성을 평가하였다. 연구결과는 첫째, 필리핀 스타트업의 기업가적 지향성은 기업의 재무적 및 비재무적 성과에 모두 긍정적인 영향을 미치는 것으로 확인되었다. 둘째, 필리핀 스타트업의 기업가적 지향성은 사회적 자본의 구조적, 인지적, 그리고 관계적 차원에서 모두 긍정적인 영향을 미치는 것으로 나타났다. 셋째, 사회적 자본의 관계적 차원이 기업가적 지향성과 기업의 재무적 및 비재무적 성과와의 관계를 모두 매개하는 것으로 확인되었다. 기업가적 지향성은 사회적 자본을 통해 스타트업의 성과에 직간접적으로 영향을 미치고 있는 것으로 확인하였다. 이러한 연구결과는 기업가적 지향성이 국내 및 미국과 같은 국가 뿐 아니라 개발도상국에서도 여전히 유효한 중요 요인임을 재확인하였다. 본 연구를 토대로 전략적 지향성의 개념과 같이 보다 통합적인 관점에서의 연구 필요성을 규명하였으며, 분석된 연구 결과를 반영하여 실무적인 시사점을 제시하였다.

Table of Contents

List of Figures

List of Tables

I. Introduction

1.1 Background of Research

Since the 2000s, the spread of IT, the Internet, and mobile has created numerous startups in the US Silicon Valley, Korea, China, and even recently in Southeast Asia. Armed with innovative technologies and ideas, startups are rapidly growing in pursuit of new opportunities in changing market flow. The successful startups have been creating tremendous economic value and becoming a new driving force in the national economy.

Globally, the cases of “Unicorns” worth more than KRW 1 trillion are appearing in the US and China. Korean has also started to see startups such as Toss and Baemin. In Southeast Asia, startups such as Grab and Lazada have exceeded KRW 1 trillion in value, and now Southeast Asian countries are pouring out various policies to encourage and support the domain venture startups.

In the case of the Philippines, the government has tried to innovate the startup ecosystem over the last three years with several issues (Isla Lipana, 2020). First of all, they reformed new rules and regulations for the innovative startup act or republic act 11337 in 2019. Second, the revised corporation code supports in the promotion of entrepreneurship for startups in the Philippines. Last, global investors such as Kohlberg Kravis Roberts & Co. L.P.(KKR), Tencent Holdings, and Ant Financial Services Group have promised to invest in the Philippines making a stronger startup ecosystem.

The basic concept of a startup can be defined as a new organization or company that develops and commercializes new business models by integrating innovative ideas and new technologies to respond to uncertain environments (Eric, 2011). However, most of these startups fail to overcome the “valley of death”, which is typically experienced in the startup life cycle. In order to determine the cause of the failure, CB Insight, a US corporate evaluation agency, surveyed 101 startups and the cause of failure. (1) No Market Need (42%), (2) Ran out of Cash (29%), (3) Not the right team (23%) were the biggest factors (CBInsights, 2019). The main factors prove that those successful startups require not only external factors such as investments, market needs but also internal capacity to lead strategically the team organization so that the entrepreneur or the start-up team can achieve visible results.

With the birth of entrepreneurs, in the meantime, scholars and researchers have been discussing the entrepreneurial spirit of founders in the process of entrepreneurship (Schumpeter, 1934; Lumpkin & Dess, 1996). The methodology for growth and performance of startups has been researched by scholars (Lumpkin & Dess, 1996; Chang & Lim, 2005). In particular, there is also a growing interest in the use of social capital to create new intellectual capital by connecting the knowledge flow of each member of the new organization as a means of entrepreneurship (Nahapiet & Ghoshal, 1998).

Nevertheless, the very few case studies verifying the hypothesis that leads to Entrepreneurial Orientation (EO), Social Capital (SC), and the performance of startups have been dealt with in Korea (Lee et al., 2019). For investors and startup founders, it was difficult to grasp various performance factors in overall startup management and administration.

This study is aiming at identifying the direct influence of entrepreneurial orientation on the performance of startups and examining social capital, as a mediator, considering internal resource combinations and environmental opportunities.

1.2 Purpose of Research

This research has two central goals: 1) to test the direct effect of entrepreneurial orientation on startup performance and 2) to explore the mediating effects of social capital in the relationship between entrepreneurial orientation and startup performance in the Philippines, investigating the main feature of social capital in the country. Through this research, it recommends new investors and partners who want to work with startups in the Philippines.

In recent studies regarding entrepreneurial orientation, it has developed into five concepts: Innovativeness, Proactiveness, Risk Taking, Autonomy, and Competitive Aggressiveness. These subordinate factors of entrepreneurial orientation have independently existed, the papers so far attempt to measure the performance of firms by adopting only a part of specific elements. However, when considering the nature of startups, in reality, each of the subfactors affects each other by the organization and its members. Thus, it has a limitation that explains the concept of entrepreneurial orientation by one factor alone.

Also, most startups are the challengers in uncertain markets. Looking to seize opportunities in the face of uncertain competition and to secure a stronghold market, startups need to take the five factors – innovation and advancement which introduces new products, risk-taking in which has to invest in uncertain environments at the expense of high cost, autonomy in which each individual and organization must practice its business or vision with the maximum assurance of independent activities, competitive aggressiveness in which each individual and organization should struggle to be more competitive. Therefore, this study aims to empirically examine the effect of startup entrepreneurial orientation on startup performance by inputting three main factors as independent variables.

In recent studies as to social capital have dealt with relational characteristics such as the range and strength of the personal network. Nahapiet & Ghoshal (1998) has widely suggested three dimensions of social capital – structural dimension as social structure, cognitive dimension as shared understandings, relational dimension as the nature and quality of relationships. With the most widely used and accepted framework, this study is going to empirically approach social capital as a mediating effect on startup performance.

Curiously, despite the rise of social capital studies in the business management area, few have attempted to address the social capital of startups (Lee et al., 2019) in Korea. For the most part, recent studies about social capital in business management have tended to center around the connection of network – relational perspectives (Tsai & Ghoshal, 1998; Yli-Renko et al., 2001; Choi, 2011). However, this study attempts to verify the hypothesis that social capital with three dimensions has a mediating effect on a startup and its growth process.

In conclusion, the purpose of this research is to verify the mediating effect of social capital along with the influence of entrepreneurial orientation and social capital on startup performance. This paper intends to investigate the following issues.

First, we define the components of entrepreneurial orientation and social capital as factors affecting startup performance based on theoretical backgrounds.

Second, this study is to analyze how the three factors – Innovativeness, Proactiveness, and Risk-Taking affect startup performance.

Third, this study is to analyze the impact of social capital consisting of three dimensions – structure, cognitive, and relationship, and how it has a mediated effect on the relationship between entrepreneurial orientation and startup performance.

At last, based on analyzing the final result of this study, this paper is going to draw suggestions on the entrepreneurial orientation and social capital, implications of startup companies that affect startup performance, suggesting future research directions. Accordingly, the present study is composed of five chapters. The contents are as followed.

Chapter 1 introduces the background, purpose, and method of research.

Chapter 2 describes the theoretical background of concepts, components, and relationships between variables based on preliminary studies on entrepreneurial orientation, social capital, and startup performance.

Chapter 3 draws a conceptual research model from the literature review discussed in Chapter 2, and sets up a research hypothesis, and sets operational definitions of variables for questionnaires.

Chapter 4 describes the collected data and the characteristics of the samples, analyzes the hypothesis of this study through statistical analysis, and interprets the results.

Chapter 5 summarizes the findings and concludes with suggestions, limitations, and directions for future research.

II. LITERATURE REVIEW

2.1 Firm Performance

In terms of firm performance, various concepts have existed from the researchers' perspective. In general, there are two ways to measure firm performance: financial performance and non-financial performance. The ways to measure and analyze the main factors of firm performance are also very diverse. From the 1980s to 1990s, financial factors such as the growth of sales, the changes in the number of employees, and the return on investment were commonly used in studies of firm performance (Brush and Vanderwerf, 1992).

However, many researchers claimed that the performance of early start-ups is inadequate to measure financial performance because it is difficult and insufficient to collect objective data from startups in the early stage. Covin and Slevin (1991) argued that firm performance in start-ups should be subjective performance indicators from the postures of founders rather than objective financial performance indicators. Maurer & Ebers (2006), who conducted the study of the relationship between social capital and firm performance with biotechnology, argued that the usual accounting factors such as revenue or firm survival are not a solution. Instead of that, they qualitatively interviewed and got factors such as patenting rate and employment growth as their assessment of performance.

Nevertheless, Chandler and Hanks (1994) emphasized that although there are many problems in measuring the performance of a start-up company, measuring both objective and subjective factors requires effective performance measurement at the expense of measurement accuracy. In addition, in the performance of start-up companies, they suggested the growth and size of their business as the most important factors and found that the growth of a startup is directly correlated with the entrepreneurial competencies.

As such, firm performance is a multidimensional construct that has been measured using a variety of indicators. In this regard, Venkatraman and Ramanujam (1986) recommended that researchers distinguish between financial and non-financial performance measures. The former indicates the achievement of the economic goals of the firm, whereas the latter captures the firm's broader operational effectiveness. Furthermore, Zahra (1996) argued that there can be a trade-off between achieving growth and profitability, suggesting that both capture distinct facets of firm performance. Accordingly, Stam et al. (2014) summarised based on prior studies that there are two different types of measures used for firm performance, namely financial measures and non-financial measures. the dimensions of startup performance are divided as followed.

Table 2-1 Division between Financial Performance and Non-Financial Performance

Type		**Performance Items**
Financial Performance	Growth	Objective or Perceived growth in sales Profit Employment market share
	Profitability	ROA (Return on Assets) ROE (Return on Equity) ROS (Return on Sales) Self-reported assessments of profitability
Non-financial Performance		Technical excellence Competitive capabilities Productivity Export performance

2.2 Entrepreneurial Orientation

2.2.1 The Concept of Entrepreneurial Orientation

For the sustainable growth of startups, the necessity of entrepreneurship has been studied extensively during the past four decades. To explain theoretically, the element of entrepreneurial orientation describes the behavior and activity of new entrepreneurship. In 1983, Miller first introduced conceptually entrepreneurial orientation (Miller, 1983). Covin and Selvin's work refines entrepreneurial orientation (Covin & Slevin, 1989). Since the mid-2000s, the work of entrepreneurial orientation has so far been studied mainly within this context, that is, as one aspect of entrepreneurship (Wales et al., 2011). This has established itself as an independent theory of the growth of SMEs (Yoon, 2015).

The concept of entrepreneurial orientation has been recognized over the past thirty years as a very important means of improving the competitiveness of firms in venture firms (Wang, 2008). As in the case of entrepreneurship of each entrepreneur, entrepreneurship in the dimension of the organization is necessary to be more successful after the organization is established. This is the definition of entrepreneurial orientation (Lumpkin & Dess, 1996; Miller, 2011).

The definition of an organization, which is holding entrepreneurial orientation, can be summarized as the company's radical innovation, progressive strategic, and risk-taking behaviors that emerge in the course of uncertain projects (Zahra & Neubaum, 1998). The stronger entrepreneurial orientation is, the more organizational capacity has which cannot be caught up and replaced by other organizations (Lumpkin & Dess, 1996).

2.2.2 The Dimensions of Entrepreneurial Orientation

The dimensions of entrepreneurial orientation proposed by Miller are the innovation, proactiveness, risk-taking (Miller, 1983). Lumpkin & Dess (1996) added the concept of autonomy and competitive aggressiveness to its components. Innovativeness can be explained by a creative behavioral tendency that draws new service or new technical processes through new ideas. Proactiveness is a tendency of firms to anticipate future opportunities and take active action responding to existing problems and services (Wiklund & Shepherd, 2005). Risk-Taking is a tendency to take bold actions that willingly input the resources of firms to new business opportunities in spite of the high-risk of failures (Rauch et al., 2009). Autonomy is a personal and organization independent behavior to achieve the vision of firms. Lastly, competitive aggressiveness is the attitude to directly challenge competitors and achieve competitive advantage (Lumpkin & Dess, 1996). The two predominant concepts are mainly suggested in the literature and can co-exist together that each approach is able to provide its own unique insights (Miller, 2011).

The past research observed that each dimension of entrepreneurial orientation is combined in various ways (Wales et al., 2011). Theoretically, some researchers argued that this research has weakened the rigor of entrepreneurial orientation research (Basso et al., 2009). Accordingly, they believe that the stabilized construct of conceptualization will give new impetus to the new creation of knowledge. Since the scope of utilization of entrepreneurial orientation is extensive, however, many researchers reinforce to suggest that alternative dimensions or factors might be needed in international entrepreneurial orientation research (Covin and Miller, 2014).

In this context, George and Marino (2011) give new concepts of entrepreneurial orientation capturing a ‘family’ of constructs. Each dimension can exist independently but it is difficult to explain holistically the concept of entrepreneurial orientation by one of three dimensions. The performance of firms is affected by shaping the level of entrepreneurial orientation, interacting with each dimension. It is important to view the independent effect of each component on startup performance, but it is more highly probable to see the entrepreneurial orientation, which consists of three dimensions, as one factor (Yoon, 2015).

Conceptually, entrepreneurial orientation is dynamic corresponding to various contexts. This can be understandable that the concept of entrepreneurial orientation can be applied to all the levels of variables – environmental, organizational, and individual (Covin and Slevin, 1991). Moreover, entrepreneurial orientation can incorporate cultural, leadership/governance, internal, external, and strategic contingency variables with firm outcomes (Miller, 2011; Wales et al, 2011). Thus, it can be concluded that entrepreneurial orientation has dynamic capabilities in terms of theoretical conceptualization and its scope of utilization. Table 1-2 shows how dimensions of entrepreneurial orientation have been used in various ways with two major performance indicators in different countries (Rauch et al., 2009)

Table 2-2 Preliminary Studies of the relationship between EO and Firm Performance

Country of Origin	Industry of firms	Dimensions	Sample Size	Performance Indicator	Reference
China	High-Tech	Innovation, Marketing Differentiation, Market Breadth, Marketing Alliance	184	Perceived Financial and Nonfinancial Performance	Haiyang, A.G. Kwaku, and Z. Yan (2000)
Korea	High-Tech	Innovativeness, Risk-Taking, Propensity, and Proactiveness	137	Perceived Financial Performance	Lee et al. (2001)
Malaysia	Mix	Innovativeness, Proactiveness, and Risk Taking	96	Perceived Financial Performance	Poon et al. (2006)
China	High-Tech	Futurity, Proactiveness, Risk Affinity, Analysis, and Defensiveness	104	Perceived Financial Performance	Tan and Tan (2005)
Belgium	Mix	Innovation, Proactiveness, and Risk-Taking	92	Perceived Financial Performance	De Clercp et al. (2003)
Korea	Technology-based firms	Innovation, Proactiveness, and Risk-Taking	277	Perceived Financial Performance and Non-financial Performance	Yoo (2001)
China and Germany	Mix	Innovation, Risk-Taking, and Proactiveness	364	Perceived Financial Performance and Non-financial Performance	Rauch et al. (2006)
Fiji	Mix	Innovation, Proactivity, Competitive Aggressiveness	71	Perceived Financial Performance	Van Gelder (1999)

Source: (Rauch et al., 2009)

2.2.3 Entrepreneurial Orientation and Firm Performance

The prior studies on Entrepreneurial Orientation have given researchers and practitioners an understanding of entrepreneurship and startup performance. However, it is still difficult to express how different organizational contexts, especially startups' link to entrepreneurial orientation. In general, Durda & Krajcick (2016) defined startups as firms that deal with a high degree of risk and uncertainty. Accordingly, early startups are required to take entrepreneurial orientation for their business. Similarly, Ghosh and Bhowmick (2014), who researched Indian R&D technology startups, pointed out that uncertainty in the startup process is a critical issue to entrepreneurs for the decision-making process when making any strategic or operational decisions in the firm. Additionally, Kee and Rahman (2018) who researched entrepreneurial orientation and startup success among Malaysian startups from a gender perspective concluded that startups need to understand the concept of entrepreneurial orientation for their future growth and survival. In various areas and perspectives, entrepreneurial orientation is the most needed in the startup process.

Many scholars have argued that the more level of entrepreneurial orientation practices results in better performance. Zeebaree and Siron (2017) researching with a sample size of 680 SMEs working in the Kurdistan Region Government in Iraq argued that entrepreneurial orientation had a positive and significant influence on competitive advantages: differentiation and low-cost as a non-finance performance factor. With 117 valid samples among beverage companies in Malaysia, Amin et al. (2016) empirically proved that entrepreneurial orientation has a significant relationship with both market orientation and firm performance, resulting in improving financial performance such as cash flow and profitability.

The positive impact of entrepreneurial orientation on firm performance is generally and widely accepted regardless of their sizes and industries to survive in the marketplace. EO gives firms a competitive advantage and is essential in firms. However, when it comes to risk-taking as one of EO dimensions, some studies have found that the results in the smaller companies and family firms have a negatively significant influence on firm performances. For example, Naldi et al. (2007), who was researching the relationship between entrepreneurial orientation and family firm performance with

696 samples including family and non-family firms, interestingly found that risk-taking in family firms is negatively related to firm performance. Although it is still arguable to generalize, risk-taking in family firms might not be working firmly in systematic and formal procedures. Unlike other elements of EO, smaller firms or early starters should be intelligent to assess the current and potential risks. Aragón-Sánchez and Sánchez-Marín (2005) claimed that engaging small firms in high-risk projects are related to the firm's sustainability when these projects long enough to get a result of their investment. Nevertheless, EO can be classified as a critical startup process that helps them to improve their performance and survive in the process (Miller, 1983).

On the basis of preceding studies from the relevant literature, it can be predicted that EO will have a positive impact on start-up performance, and the study proposes the following research hypotheses.

Hypothesis 1.1. There is a positive relationship between Entrepreneurial Orientation and Financial Performance.

Hypothesis 1.2. There is a positive relationship between Entrepreneurial Orientation and Non-Financial Performance.

2.3 Social Capital

2.3.1 The Concept of Social Capital

Since the first appearance of social capital in 1916, several studies in social science have been historically conducted in numerous perspectives and arguments. Intermittently, during the 1960s-1980s, some sociologists and political scientists have used and identified the term (Jacobs, 1961; Granovetter, 1973; Bourdieu, 1986). Adler and Kwon (2002) summarized that Social capital as the main concept has been used in various areas of research including economic development and organization studies.

The concept of social capital has been historically developed by numerous scholars. Coleman (1988) suggests the concept of social capital with three forms – obligations and expectations, information channels, and social norms. The effect of social capital arises when relationships between people in the network lead to productive behavior. Putnam (1993) claimed that the successful accumulation of social capital results in developing a well-functioning economic system and a high level of political integration. He has three components of social capital: obligations and norms, trust, and voluntary associations. According to Nahapiet & Ghoshal (1998), social capital can be defined as the sum of all practical and potential resources available inherent in the network. Tsai and Ghoshal (1998) defined social capital as encompassing social relationships, trust relationships, and a value system that facilitates the behavior of individuals located within these contexts. They have proved the importance of social capital within the firms how each element of social capital has affected its innovative performance and interacted with the existing resources from the firm. Adler and Kwon (2002) defined that social capital is an exclusive dimension which derives from a specific network and can be used by the participants in the network.

Previous researches on social capital have suggested that a person can acquire a higher quality of information by utilizing his or her social network. Depending on the elements of social capital held by the startup team, the quality of information that can be obtained in the startup process varies, implying that this could affect firm performance. Empirically, the positive relationship between the social capital of founder and firm performance has been proved in recent studies (Maurer

& Ebers, 2006; Smith et al., 2017). Stam et al. (2014), who had conducted a meta-analysis of social capital of entrepreneurs and small firm performance with 61 samples, found that the social capital of entrepreneurs constitutes a key asset for small firms, utilizing social capital for them to access the necessary resources with lower price than market price and leading them to find new business opportunities.

Based on prior research such as Nahapiet and Ghoshal (1998), three dimensions of social capital: structural, relational, and cognitive dimensions were set up and measured by conceptualizing the combination and strength of the startup's network, trust among members within the startup, and shared vision and goals between members.

2.3.2 The Dimensions of Social Capital

Stam et al. (2014) have collected the findings from 61 studies, conducting a meta-analysis of social capital. To assess the generalizability of results regarding the performance effects of entrepreneurs' social capital, they tried to synthesize the findings from various studies. They found out that many studies have differently labeled in identifying social capital. Scholten (2006) suggested the division of social capital since Social capital has been differently conceptualized in many articles by setting the varying defining properties of social capital to three dimensions: the structural, relational, and cognitive or contingent dimension. The relevant table 2-2 is as followed.

Nahapiet and Ghoshal (1998) subdivided three dimensions of social into nine properties. When explaining the relationship of three dimensions, the authors assumed that the impact of each dimension exists independently. They recognized, however, that both the dimensions and the several properties of social capital could have interacted in complex ways when creating and exchanging new information or knowledge. The definition of nine properties in three dimensions of social capital is explained as followed.

Table 2-3 shows a summary of different properties in each dimension of social capital. In short, the structural dimension refers to the strength of ties between members in the network to which an individual belongs. At the structural dimension of social capital, the stronger network attributes, the more accessible the members of the organization to the knowledge and information they need, which improves the quality and timeliness of knowledge and information (Adler & Kwon, 2002).

In this dimension, new information and knowledge can be creatively drawn from the exchange and sharing of resources between networks (Nahapiet & Ghoshal, 1998). The cognitive dimension of social capital represents the bond and homogeneity that enables the shared language, codes, and narratives between members to have shared value and vision. Since this dimension develops mutual understanding capacity between members (Scholten, 2006), a better understanding between actors encourages more the use of social capital on accessing each other's information and resources (Maurer & Ebers, 2006). The relational dimension is based on qualitative relationships with members such as inherent trust and norms that effectively affect resource acquisition. This dimension is oriented to the relational aspects which can either facilitate the use of social capital. It can reduce surveillance and transaction cost via trust and norms and serve as the basis for the exchange of information and combination, and cooperative activities (Putnam, 1993). It has a positive effect on the exchange, combination, and accumulation of resources with other dimensions of social capital (Tsai & Ghoshal, 1998).

Lee & Yoon (2009) recognized social capital as an interaction that enables various activities inside and outside the market and played a role of resource in connecting between firms and market information. In this perspective, the dimensions of social capital are divided as followed.

Table 2-3 Dimensions of social capital and corresponding benefits

Dimension	Ends of the continuum	Benefits
Structural	Redundancy (closure argument)	Trus and enhanced communication Reputation Continuity of the linkage
	Non-redundancy (structural hole argument)	Information and Resources Time advantage Referrals
Relational	Strong-tie	Reliable Contacts Transfer of complex knowledge
	Weak-tie	Opportunities and unique resources
Contingent	Content of tie	Effectiveness of network structure or strength of tie depends on the content of discussion and context

Source: (Scholten, 2006, p. 46)

Table 2-4 Definitions of the Dimensions of Social Capital

Dimension	Definition
Structural Dimension	Network type or characteristics between actors
Cognitive Dimension	Communication for information exchange, problem resolution, and conflict management
Relational Dimension	The extent to which members can contribute to profit or value creation, trust as a key concept can reduce transaction costs.

Table 2-5 Dimensions of Social Capital and their defining properties

Dimension	Summary of Properties
Structural Dimension	Network ties - This property can help a person to access valuable resources and play a role as an information channel in reducing the unnecessary amount of time and investment for he/her to gather important information for action. Network Configuration - While network ties support a channel for important information transmission, the network configuration of ties develops an important relation that may impact on shaping intellectual capital. Appropriable Organization - This property can provide a potential network that can access to the right people and resources, including information and knowledge.
Cognitive Dimension	Shared language and codes - This property gives a boost to a person boost up access to the right people and their information. The shared language may support a common conceptual mechanism for evaluating the benefits of information, enhancing combination capability within firms. Shared narratives - The narrative mode related to myths, stores, and metaphors provide powerful means that can facilitate the combining of both imaginative and literal observations and cognitions. In doing so, the shared narratives can provide a new perspective and interpretation of information between members.
Relational Dimension	Trust - The higher in trust, the more people are willing to engage in social exchange and cooperative interactions. Trust can help people to open a mind up access to the right people for the exchange of information and increase the expectation of value. Once trust becomes a generalized norm of cooperation over time, this trust will help to solve problems of cooperation and coordination within firms. Norms - This property enables to establish a strong foundation for the creation of intellectual capital and influences on exchanging process for the knowledge and information, including a willingness to value and respond to diversity and even a tolerance of failure. This can be lead to “groupthink” among members. Obligations and expectations - This property is a commitment or duty to execute some activity to access to parties for exchanging and combining knowledge. The formal, professional, and personal obligations can provide involved and cooperative research and development projects between different firms. Identification - This property is the process by which each person sees himself or herself as one with another person or group of people. Identification acts as a resource affecting both the expectation of value to be collected from the combination and exchange of intellectual capital.

2.3.3. Entrepreneurial Orientation and Social Capital

Previous literature has used entrepreneurial orientation and social capital as independent variables without considering their interrelationships. Stam and Elfring (2008), however, summarised that a configurational perspective on entrepreneurial orientation (Wiklund & Shepherd, 2005) and social capital have been conducted in recent studies, interacting with other components and giving a better understanding of their effects (Mehra et al. 2006). Anderson et al. (2007) concluded in their research that social relations, interaction, and networks are important in the entrepreneurship process if the development of startups is perceived as a socio-economic process.

De Clercq et al. (2013) found in their research that firms with higher levels of internal social capital enhance the higher levels of entrepreneurial orientation. Most startups that lack resources and competencies are highly dependent on internal members' knowledge of information capabilities to acquire new opportunities and create value for use (Moon and Lee, 2016). Social Capital influences the firm's ability to combine knowledge across various functional teams that finally affects its ability to find new opportunities successfully (De Clercq et al., 2010). Social Capital provides value to the actors immerse in it, allowing taking advantage of the resources established in actors' relationships (Bourdieu, 1986) and get a better competitive advantage (Tsai & Ghoshal, 1998).

Reviewing the relationship between entrepreneurial orientation and social capital, entrepreneurial orientation facilitates to create new social capital as a channel achieving new knowledge and technology (Marino et al., 2002). In the process of seeking innovative opportunities, the number of the network is increasing and the cognitive system that transits the necessary resource is shared in the network. For instance, Lee et al., (2001) researched 137 Korean technological startup companies that the higher entrepreneurial-oriented companies utilizing various linkages have more abilities to gain valuable resources and economic opportunities through their social network and then create more value for customers and businesses. It is difficult for non-entrepreneurial oriented companies to obtain precious resources from their social capital. Accordingly, as startups strengthen their entrepreneurial orientation, which is innovative decision-making and proactive action to predict new market demands and opportunities over competitors, the scope and content of a startup's network can be rich, integrating with an internal and external resource. The network interaction will strengthen the cognitive system of startup members and accumulate the trust and norms between them.

Based on preceding studies from the relevant literature, it can be predicted that EO will have a positive impact on Social Capital, and the study proposes the following research hypotheses.

Hypothesis 2.1. There is a positive relationship between Entrepreneurial Orientation and the structural dimension of social capital.

Hypothesis 2.2. There is a positive relationship between Entrepreneurial Orientation and the cognitive dimension of social capital.

Hypothesis 2.3. There is a positive relationship between Entrepreneurial Orientation and the relational dimension of social capital.

2.3.4. The Mediating Effect of Social Capital on Firm Performance

Social capital can be categorized into external and internal capital (Adler & Kwon, 2002). The external social capital is defined as groups such as suppliers, customers, and professional associations. These resources are rooted in the concepts of trust, loyalty, and referrals. Internal social capital is more related to groups such as friends, families, business partners, and employees. These are rooted in connections with trust, support, and even business strategic advice.

Nahapiet and Ghoshal (1998) suggested that internal social capital consists of three dimensions: structural, relational, and cognitive dimensions. Some authors dealt with three dimensions of internal social capital independently (Nahapiet & Ghoshal, 1998) whereas other studies argued that three dimensions are interacted (Tsai & Ghoshal, 1998).

A few previous works of literature have researched the role of social capital in firm performance. Commonly, depending on the scope and range of social capital utilized by a startup, social capital has affected the relationship between entrepreneurial orientation and firm performance. For example, Jang (2006) researched 115 technology-based ventures in Korea on how the human capital of the founder and social capital of the firm from the founder influences the firm performance in the circumstance of unsystematic risks such as the national financial crisis and epidemic are influenced to survive.

The study was found that the founder utilizing social capital and human capital was significantly related to firm survival from the unexpected crisis from external environments. Stam and Elfring (2008) researched 90 new ventures in the emerging open-source software how the configuration of a founding team's social capital shapes the relationship between entrepreneurial orientation and firm performance. The study was found that the firms with few structural dimensions of social capital as network ties and centrality weakened the relationship between entrepreneurial orientation and firm performance as expected. In the empirical study using a sample of 142 manufacturing and service companies, it was founded that the internal social capital has a greater impact on the technological and market situation of the enterprise than the external capital (Cuevas-Rodriguez et al., 2014). Besides, recent empirical evidence on the impact of social capital on firm performance tends to focus mainly on the culture of a firm, internal social capital than the external social capital (Wang & Steiner, 2019). In the entrepreneurship literature, some researchers analyze the impact of social capital as a whole on firm performance. However, a few studies have been conducted on the role of each dimension of social capital (Fornoni et al., 2012) on startup performance. The following discussions will explain the pros and cons of the relationship between internal social capital and startup performance.

Hernadez-Carrion et al. (2016) examined the impact of entrepreneurs' social capital on the economic performance of 951 surveyed small businesses and found that economic performance is influenced more by entrepreneurs' own professional and institutional network resources than by other network resources. Saha and Benerjee (2015) survey small businesses in West Bengal and found that the impact of social capital on firm performance is significantly greater in firms engaged in formal and informal networking in contrast to firms embedded only in the informal network. Based on the empirical studies, it can be expected that the social capital in startups will play a role as a mediate variable between entrepreneurial orientation and firm performance. The study proposes the following research hypotheses.

Hypothesis 3-1. The structural dimension of social capital in startups will mediate the relationship between entrepreneurial orientation and financial performance.

Hypothesis 3-2. The cognitive dimension of social capital in startups will mediate the relationship between entrepreneurial orientation and financial performance.

Hypothesis 3-3. The relational dimension of social capital in startups will mediate the relationship between entrepreneurial orientation and financial performance.

Hypothesis 3-4. The structural dimension of social capital in startups will mediate the relationship between entrepreneurial orientation and non-financial firm performance.

Hypothesis 3-5. The cognitive dimension of social capital in startups will mediate the relationship between entrepreneurial orientation and non-financial performance.

Hypothesis 3-6. The relational dimension of social capital in startups will mediate the relationship between entrepreneurial orientation and non-financial performance.

III. METHODOLOGY

3.1 Research Method

The purpose of this study is to examine the effect of entrepreneurial orientation and social capital on firm performance and to empirically prove the mediating effect of social capital on the relationship between entrepreneurial orientation and firm performance.

First, the subordinate factors of entrepreneurial orientation – innovativeness, proactiveness, and risk-taking are used as independent variables that are the most frequently used in the related meta-study and considered to have high feasibility (Yoon, 2014; Rauch et al., 2009). Yoon (2015) stated that the characteristics of each factor are related and identified as a whole concept that cannot be explained separately. Accordingly, three subordinate factors of entrepreneurial orientation will be applied as the single dimension out of the average sum of each factor on how entrepreneurial orientation affects social capital and firm performance.

Second, three sub-factors suggested by Nahapiet & Ghoshal (1998) will be operationally defined as an organizational perspective rather than the founder's personal view. The structural dimension will be defined as a concept of connectivity and density related to business and non-business inside the team (Nahapiet & Ghoshal, 1998). The cognitive dimension will be defined as a concept of the degree of employee involvement with shared vision and goals inside the team (Adler & Kwon, 2002). The relational dimension will be defined as a concept of trust and reciprocity among team members. Each dimension will be utilized and analyzed as a mediator that affects the relationship between entrepreneurial orientation and firm performance (Tsai & Ghoshal, 1998).

For this empirical study, this paper sets up and presents a conceptual research model as shown in Figure 3-1.

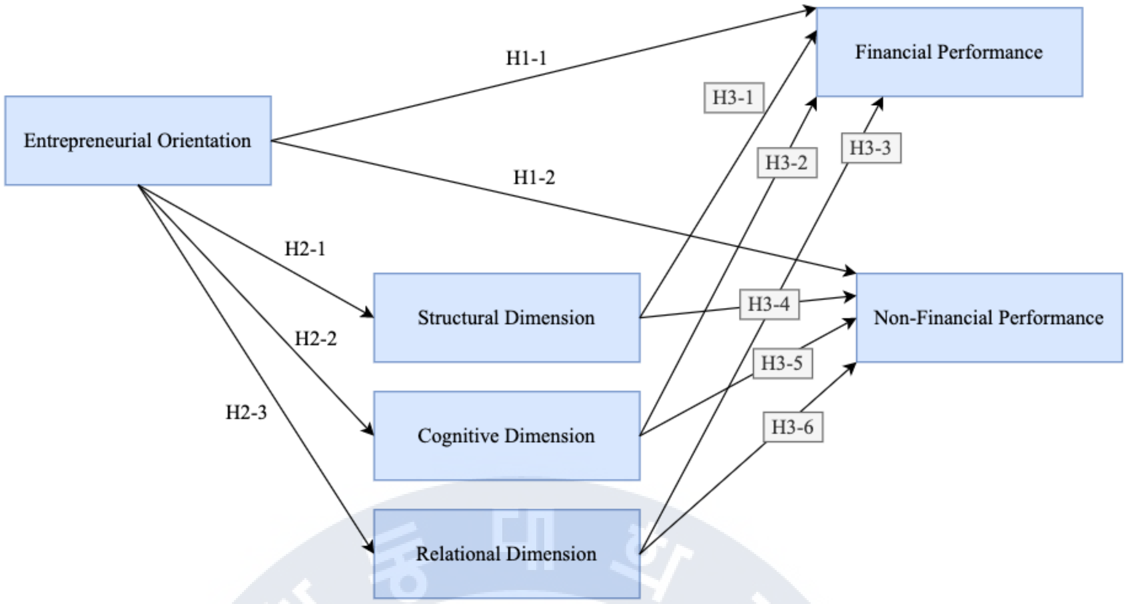

Figure 3-1 Research Design

The present study presented the theoretical background and research direction through the method of literature research. Empirical research is conducted to verify the derived conceptual research model and the theory of this study using the statistical analysis method.

In the theoretical background, this paper tried to look at the flow and concepts of research on entrepreneurial orientation, social capital, and startup performance, which organized the components of each variable based on the literature from various studies. Also, the conceptual research model and research theory were established through the consideration of the relationship between variables to provide the basis for empirical research. (Chang & Lim, 2005)

In the empirical study, after setting operational definitions of the variables, the main variables were selected based on the foregoing theoretical considerations. These variables are used to make questionnaires so that the hypothesis was verified and analyzed. The questions in the questionnaire were organized from the preceding study, consisting of nine questions of entrepreneurial orientation, nine questions of social capital, six questions of startup performance, and five questions of control variables, and five questions of demographic characteristics.

3.2. Operational Definition of variables and Construction of Survey

3.2.1. Entrepreneurial Orientation - Independent Variable

In the case of start-ups, it is natural for startups to make a challenging decision in uncertain markets as considering a few available resources compared to their existing competitors. The entrepreneurial orientation of the startups is an essential element in the start-up process of exploring the market with new ideas and technologies. Therefore, this study viewed entrepreneurial orientation as the tendency of start-up teams to proactively create new markets by using innovative technologies in response to various difficulties in the start-up process.

Three dimensions of entrepreneurial orientation – innovativeness, proactiveness, risk-taking were measured using the scale suggested by Covin and Slevin (1989), which is an acceptable and validated nine-item, seven-point scale (Santarelli & Tran, 2013; Yoon, 2015). While Lumpkin & Dess (1996) argued that three dimensions of EO can be measured as independent variance, most empirical studies have conducted that the three dimensions of EO are of 'equal importance in explaining business performance' (Rauch et al., 2009).

As such, a modified version of Covin and Slevin (1989) nine item-scale is employed in a single statement format, asking the respondents to indicate their perception on a five-point Likert-type scale, ranging from 'strongly agree' to 'strongly disagree'. The three factors are considered as one holistic concept (Yoon, 2014). Thus, the statistical significance will be verified by applying the average sum of each factor.

Table 3-1 Operational Definition of Entrepreneurial Orientation

Variables	Definition of Variable	Related Studies
Innovativeness	The will of a startup team engaging in creative experimentation through the new products and services.	Covin & Slevin, 1989, Rauch et al., 2009
Proactiveness	The action of the startup team seeking a new opportunity and forward-looking perspective with new products and services and acting in anticipation of future demand.	
Risk-taking	The mindset of a startup team involving in bold actions by adventuring into committing significant resources to the team in uncertain environments.	

3.2.2. Social Capital - Independent Variable

The social capital of a startup is an important source that can create intangible resources in the startup process. It can be an environmental feature that can affect the founder. This is an exclusive resource that can be accessed or used only by the startup members participating in the startup to achieve their business performance goals (Nahapiet & Ghoshal, 1998; Adler & Kwon, 2002).

Based on previous researches, three dimensions of social capital are divided into three groups as an independent factor. First, the structural dimension of social capital is defined as the social interaction, network ties, and knowledge shared between members in a startup that characterizes the frequency and density related to business and non-business between startup members. Second, the cognitive dimension of social capital is defined as the cohesion that features the shared visions and immersion on it between startup members at each department in the startup team. Last, the relational dimension of social capital is defined as the trust of a startup team that applies trust and norms between members in the same network (Tsai & Ghoshal, 1998).

Unlike the entrepreneurial orientation, many researchers have different views in social capital, depending on contexts – individual, organizational, and society (Claridge, 2018). As such, adopting questions from previous studies (Tsai & Ghoshal, 1998; Yli-Renko et al., 2001; De Clercq et al., 2010) is employed in questionnaire development. Each dimension is treated as each variable that will be measured based on the survey and the items were rated on a 5 point Likert scale (1 = strongly agree, 5 = strongly disagree). The operational definition item of each variable is designed as based on the literature review shown in the table below.

Table 3-2 Operational Definition of Social Capital

Variables	Definition of Variable	Related Studies
Structural Dimension	The social interaction, network ties, and knowledge shared between members in a startup	Tsai & Ghoshal (1998) Yil-Renko et al. (2001) Chiu et al. (2006) De Clercp et al. (2009)
Cognitive Dimension	The cohesive power between startup members	
Relational Dimension	The trust and norm between startup members	

3.2.3. Firm Performance - Dependent Variable

In the methodology of measuring firm performance, many researchers have traditionally utilized a financial measurement method. When applying it to a startup company, it may not be desirable to measure objective financial performance indicators (Covin and Slevin, 1991). Alternatively, Chandler and Hanks (1994) suggested that measuring both objective and subjective performance factors are ideal and more effective to evaluate startups.

Stam et al. (2014) who had meta-analysis about the social capital of entrepreneurs and small firm performance indicated that the social capital performance was stronger for measures of nonfinancial performance than financial performance. This finding implies that if studies only focus on the financial performance of a startup, then they may miss the true value of social capital.

Accordingly, firm performance will be divided into financial and non-financial performance. First, financial performance is defined as the startup's qualitative growth. In this case, the profitability indicators such as ROI, ROE, and ROA are excluded in this survey. Rather, this study focuses on startup growth in sales, the number of employment, marketing budget, and so on during the recent six months in perceived view. Second, non-financial performance is defined as growth potential. In this performance, technical excellence, competitive capabilities, and evaluation from other reliable organizations.

Considering the construct of questionnaires, this study is based on the division of Table 2-5, "Division between Financial Performance and Non-Financial Performance". From this basis, this study is adopted from various studies. For financial performance, the questions in previous literature (Lee, 2012; Zahra, 1996) are adopted and modified. For non-financial performance, the questions in related studies (Kim et al., 2014; Cooper & Kleinschmidt, 1995) are modified. Like independent variables, a 5 point Likert scale (1 = strongly agree, 5 = strongly disagree) will be applied to the items to be rated. The summary of the operational definition is shown in the table below.

Table 3-3 Operational Definition of Social Capital

Variables	Definition of Variable	Related Studies
Perceived Financial Performance	Startup's qualitative growth that team members are numerically perceived.	Kim et al. (2014) Lee (2012) Zahra (1996) Chandler & Hanks (1994)
Perceived Non-Financial Performance	Startup's growth potential with technical excellence and external evaluation.	

3.2.4. Control variables

Younger and smaller firms are likely to face more difficult challenges in exploring opportunities because of their small resources. In most cases, the founder's intrinsic capacity is the main driving force in the early stage of a startup. Regardless of independent variables, firm performance will be affected by the founder's capacity, firm size, and firm age (Jang, 2006; Stam and Elfring, 2008). These variables are used as control variables. First, firm age is calculated via logarithm of the number of years since a firm was founded. Second, firm size is also calculated through the logarithm of the total members of employees including founders. Last, the respondent's prior work experience is controlled as a dummy variable. When defining prior work experience, Florin et al., (2003) divided human resources into industry experience, startup experience, and venture capitalist directorships, before founding the current venture. In this study, each experience is treated as a dummy variable (0 = No experience, 1 = Experience).

3.3. PLS-SEM

3.3.1. PLS-SEM and CB-SEM

Structural Equation Modeling (SEM) is divided into two methods Covariance-Based SEM (CB-SEM) and Partial Least Squares SEM (PLS-SEM). In general, the goal of CB-SEM usage is theory testing and theory confirmation, accordingly, confirmatory factor analysis (CFA) is used in this method. On the other hand, the goal of PLS-SEM is predicting key target constructs or identifying key driver constructs, thus exploratory factor analysis (EFA) is used in this method (Hair et al., 2016). The comparison is described in Table 3-4. Based on the comparison, CB-SEM is an appropriate analysis for larger data that is parametric which does assume normal distribution to confirm a theory. On the other hand, PLS-SEM is an appropriate analysis for small data and complex path models since the approach of this technique is non-parametric which does not assume normal distribution to figure out the key construct and predict.

Since the main objective of this research is to explore, find key features of social capital, PLS-SEM will be used as a method.

Table 3-4 Comparison between CB-SEM and PLS-SEM

Criteria	**CB-SEM**	**PLS-SEM**
Objective	Theory Testing	Prediction oriented
Required Sample Size	200~800	30~100
Distribution Assumptions	Parametric	Non-Parametric
Model Complexity	Less than 100 indicator variables	Large Models Possible
Statistical tests for parameter estimates	Assumptions must be met	Inference requires Bootstrapping
Measurement Model	Typically only Reflective Indicators	Formative and Reflective Indicators
Goodness-of-fit measures	Many	None

(Source: Shin, 2018, p. 29)

Some researchers often believe that the PLS-SEM approach allows them to use a very small sample to take great results representing the effects that can apply in a big population. However, there is no magic analysis technique including PLS-SEM with a very small sample size. However, two early studies evaluated the performance of PLS-SEM with small sample size and proved that PLS-SEM performed well (e.g., Chin & Newsted, 1999; Hui & Wold, 1982). However, a problem with small size often occurs when using CB-SEM on complex models with a small sample size. By then, PLS-SEM can be an alternative to CB-SEM (Henseler et al., 2014).

Since the number of startups in the Philippines is relatively smaller than advanced countries and the limited envorionment to collect data in the era of Covid-19, the researcher decided to use the PLS-SEM technique in this research.

3.3.2. Data Analysis Procedure

For data analysis, both R 4.0.2 (R Core Team, 2020) and SmartPLS 3.3.2 (Ringle, Wende, & Becker, 2015) were used. R is an open-source programming language for statistics. R was utilized for data cleansing, descriptive statistics, and exploratory factor analysis. SmartPLS tested the hypothesized relationships among entrepreneurial orientation, social capital, and start-up performance.

The process of data analysis is as followed.

First, the researcher collects data from the founder or employees like the director or manager in each startup.

Second, the researcher cleans and filter data removing irrelevant for data analysis

Third, the researcher tests internal consistency reliability, convergent validity, and discriminant validity of indicators and constructs are valid or not.

Fourth, the researcher evaluates the structural model with collinearity issues, coefficient of determination, effect size, predictive relevance, and the significance of path coefficients.

Last, the researcher finally tests the hypotheses based on results generated by SmartPLS.

The acceptance criteria in each process will be explained in Chapter 4.

3.4. Sampling

Sample size recommendations in PLS-SEM are essentially built on the properties of OLS regression. Chin (1998) and Barclays et al. (1995) proposed 10 times rules in determining the minimum sample size. However, if a structural model is constructed solely from a reflected measurement model, it should be greater than 10 times the maximum number of paths leading to a particular latent variable. Table 3-5 shows the sample size requirements corresponding to R^2 values of at least 0.25 (with a 5% probability of error) with a statistical power of 80%.

Table 3-5 Determination of Sample Size

Maximum Number of Arrows pointing at a Construct (Number of independent variables)	Significance Level											
	10%				5%				1%			
	Minimum R^2				Minimum R^2				Minimum R^2			
	0.10	0.25	0.50	0.75	0.10	0.25	0.50	0.75	0.10	0.25	0.50	0.75
2	72	26	11	7	90	33	14	8	130	47	19	10
3	83	30	13	8	103	37	16	9	145	53	22	12
4	92	34	15	9	113	41	18	11	158	58	24	14
5	99	37	17	10	122	45	20	12	169	62	26	15
6	106	40	18	12	130	48	21	13	179	66	28	16
7	122	42	20	13	137	51	23	14	188	69	30	18
8	118	45	21	14	144	54	24	15	196	73	32	19
9	124	47	22	15	150	56	26	16	204	76	34	20
10	129	49	24	16	156	59	27	18	212	79	35	21

(Source: Cohen, 1992)

Although these 10 times rules provide an overview of the small sample size requirements, it may be a good idea to use the G*Power program (http://www.gpower.hhu.de) to conduct a statistical power analysis and determine the sample size as a result to determine better minimum sample size. Figure 3-2 represents how the minimum sample size is necessary to collect data from the population.

With support from Pinoy Seoul Media Enterprise in Korea and FastJobs Philippines Inc., this study surveyed start-ups registered in the Philippines. For 60 days from 1 June to 30 July 2020, the survey was conducted by founders or employees of start-ups. A total of 241 startups in the Philippines were contacted by a support agency. Among them, 108 surveys were collected by web questionnaires with 44.8% response rate. Finally, 93 responses of 108 were used for analysis, excluding 15 responses because of the missing values and duplicated responses. The collected sample size is met with the minimum sample size of 85 given in Figure 3-2.

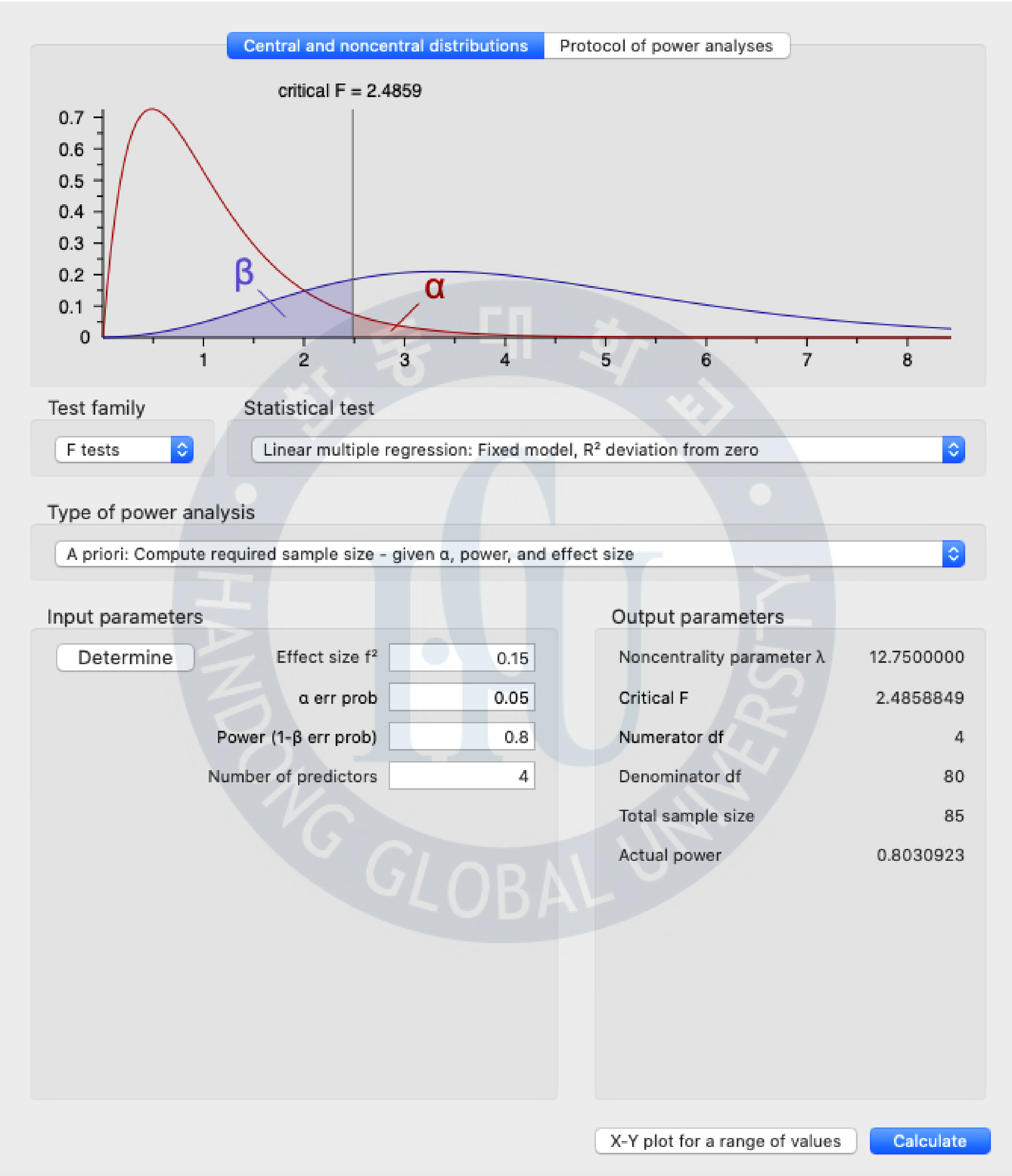

Figure 3-2 Suggestion of Minimum Sample Size using G*Power Program

Table 3-6 Measurement of Variables

Variables		Measurement Metric		Items	Reference
Entrepre-neurial Orientation	Innovativeness	EI_1	The strong emphasis on R&D, technological leadership, and innovations	9	Covin & Slevin, 1989, Rauch et al., 2009
		EI_2	New lines of products or services in the past 1-3 years		
		EI_3	Dramatic Changes in product or service lines		
	Proactiveness	EP_1	Initiative actions which follow competitors to respond		
		EP_2	The first mover to introduce new products/services, techniques		
		EP_3	Adoption of competitive and undo-the-competitors' posture		
	Risk-Taking	ER_1	A strong proclivity for high-risk projects		
		ER_2	Bold, wide-ranging acts are necessary to achieve the firm's objectives		
		ER_3	Maximizing the probability of exploiting potential opportunities		
Social Capital	Structural Dimension	SS_1	The maintenance of close social interaction with members	9	Tsai & Ghoshal (1998) Yil-Renko et al. (2001) Chiu, et a., (2006) De Clercp et al. (2009)
		SS_2	The social interaction with some member on a personal level		
		SS_3	The network−ties to commit between members		
	Cognitive Dimension	SC_1	The shared same ambitions and vision at work		
		SC_2	The enthusiastic pursuit of the collective goals and mission		
		SC_3	The valuable knowledge shared with other members		
	Relational Dimension	SR_1	The promise between members		
		SR_2	The trustworthy between members		
		SR_3	The personal belief that gets help from other colleagues		
Firm Performance	Financial Performance	F1	The perceived growth in market share	3	Kim (2014) Lee (2012) Zahra (1996) Chandler & Hanks (1994)
		F2	The perceived change in cash flow		
		F3	The perceived sales growth		
	Non-Financial Performance	NF1	The goal achievement by internal evaluation	3	
		NF2	The growth potential evaluation by the competitors and venture capitalist		
		NF3	The registration of patent and innovation		
Control Variables	Firm Age	Age	Firm Age	1	Jang, 2006 Stam & Elfring, 2008 Florin, Lubatkin, & Schulze, 2003
	Firm Size	Scale	The number of employees	1	
	Respondent's Work Experience	WE1	Respondent's prior work experience with the startup industry	3	
		WE2	Respondent's prior work experience with venture capitalist directorship		
		WE3	Respondent's work experience with the same industry		

Chapter 4. Results of Research

4.1. Demographics of Respondents

Table 4-1 shows that a frequency was conducted to determine the demographic distribution of 93 respondents measured in this study.

Table 4-1 Demographic Distribution of Survey Respondents (N=93)

Category		Frequency	Percent (%)
Gender	Female	56	60.0
	Male	37	40.0
Position	Core Employee	63	68.0
	Founder of Company	30	32.0
Age	Younger than 30	49	53.0
	30-39	34	37.0
	40-49	8	8.6
	50 or Above	2	2.2
Education	High School	6	6.0
	Undergraduate School	64	69.0
	Graduate School	23	25.0
Business Area	E-Commerce	16	17.0
	Education	4	4.3
	Energy	1	1.1
	Enterprise Services	4	4.3
	Fintech	9	9.7
	Logistics	5	5.4
	Media and Entertainment	6	6.5
	Medical and Healthcare	1	1.1
	Online to Offline Commerce	2	2.2
	Others	39	42.0
	Real Estate and Household	1	1.1
	Transportation/Automotive	4	4.3
	Travel	1	1.1

The survey results were presented with a total of 93 respondents. Among respondents, 56(60.0%) were female and 37(40.0%) were male. In the current position, 63(68.0%) were an employee as manager or director and 30(32.0%) were the founder of the company. In age, 49(53.0%)

were younger than 30, 34(36.5%) in 30-39, 8(8.6%) in 40-49, 2(2.2%) in 50 or above. In education, 64(69.0%) were from undergraduate school, 23(25.0%) were from graduate school, and 6(6.0%) were from high school.

The categories of business fields were adopted from the reports (Isla Lipana, 2020). The comparison of the two surveys showed in Table 4-2.

Table 4-2 Survey Comparison between Current Research and PwC Research

Category	**Current Research (N=93)**		**PwC Research* (N=111)**	
	Percent	Rank	Percent	Rank
E-Commerce	17.0 %	2	11 %	4
Education	4.3 %	6	9 %	5
Energy	1.1 %	11	1 %	12
Enterprise Services	4.3 %	6	15 %	2
Fintech	9.7 %	3	12 %	3
Logistics	5.4 %	5	3 %	10
Media and Entertainment	6.5 %	4	6 %	7
Medical & Healthcare Real Estate & Household	2.2 %	9	4 %	9
Online to Offline Commerce	2.2 %	9	5 %	8
Others	42.0 %	1	21 %	1
Transportation & Automotive	4.3 %	6	8 %	6
Travel	1.1 %	11	2 %	11

Source: (*Isla Lipana, 2020, p. 30)

In the business field, 39(42%) are from others, which is similar to the survey results from the PwC consulting firm. 16(17%) are currently working in the e-commerce area, 9(9.7%) are working at Fintech, 6(6.5%) are working at Media and Entertainment, 5(5.4%) are working at Logistics, 4(4.3%) are working at Transportation/Automotive, Enterprise Services, and Education field. 2(2.2%) are working at Online to Offline Commerce, Medical & Healthcare, and Real Estate & Household. 1(1.1%) in Energy and Travel answered the survey. To sum up, IT tech-related with business fields, such as e-commerce and fintech accounts for high ranks in this research as the PwC report says. This can say that when it comes to startup, most startups are IT-driven companies in the Philippines. But, one disadvantage of this research compared with PwC is that the proportion of others (42%) are much higher than PwC (21%). Nonetheless, the demographic of respondents in this research can be similar to it in PwC.

Lastly, the experience of a startup, venture capital, and industry are also surveyed as followed in Table 4-3.

Table 4-3 Frequency Analysis of Control Variables

Category		Frequency	Percent (%)
Firm Age	Less than 2 years	28	30.0
	3-4 years	27	29.0
	5 years above	38	41.0
Firm Size	Less than 5 members	15	16.1
	5-9 members	16	17.2
	10-14 members	16	17.2
	Above 15 members	46	49.5
Startup Experience	One time	17	18.2
	Two times	5	5.4
	More than three times	9	9.7
	No Experience	62	66.7
Venture Capital Experience	Yes	19	20.4
	No	74	79.6
Same Industrial Experience	Yes	44	47.3
	No	49	52.7

In the firm age, 38(41.0%) are working at the firms over five years, 28(30.0%) are working at the firms since 2018, and 27(29.0%) are working at the firms since 2016. In the firm size, 46(49.5%) are working at the firms where above 15 employees are working, 16(17.2%) are working at the firms both with 10-14 members and with 5-9 members. 15(16.1%) are working at smaller firms with less than 5 members. 31(33.3%) of respondents had experienced working in a startup company more than one time, whereas 62(66.7%) of respondents had no experience before joining the current company. In venture capital experience, 74(79.6 %) had no experience, whereas 19(20.4%) had experienced. Finally, 49(52.7%) of respondents were working from a different business field, whereas 44(47.3%) were working in the same industry before.

4.2. Descriptive Statistics

Figure 4-1 and Figure 4-2 show the distribution of each item using a bar graph. Questionnaires about independence consist of entrepreneurial orientation which contains EI (Innovativeness), EP (Proactiveness), and ER (Risk-Taking), and social capital which includes SS (Structural Dimension), SR (Relational Dimension), and SC (Cognitive Dimension). Likewise, questionnaires about dependent variables consist of financial and non-financial performance.

In entrepreneurial orientation, respondents in the Filipino startups have a tendency of taking risk-taking and innovativeness rather than proactiveness. In social capital, respondents answered positively relational dimension is an important factor relatively rather than cognitive dimension and structural dimension. Figure 4-1 presents that the relational dimension among independent variables takes the highest point from respondents, in particular, 87% points to strongly agree with the question "when confronted with solving any kind of problems, I believe that members in my team would help me if I need". On the other hand, proactiveness is a relatively low point, in particular, 68% points to strongly agree with the question "in dealing with its competitors, my firm changes in product or service lines have usually been quite dramatic." Unlike independent variables, the dependent variables in financial and non-financial performance are evenly distributed.

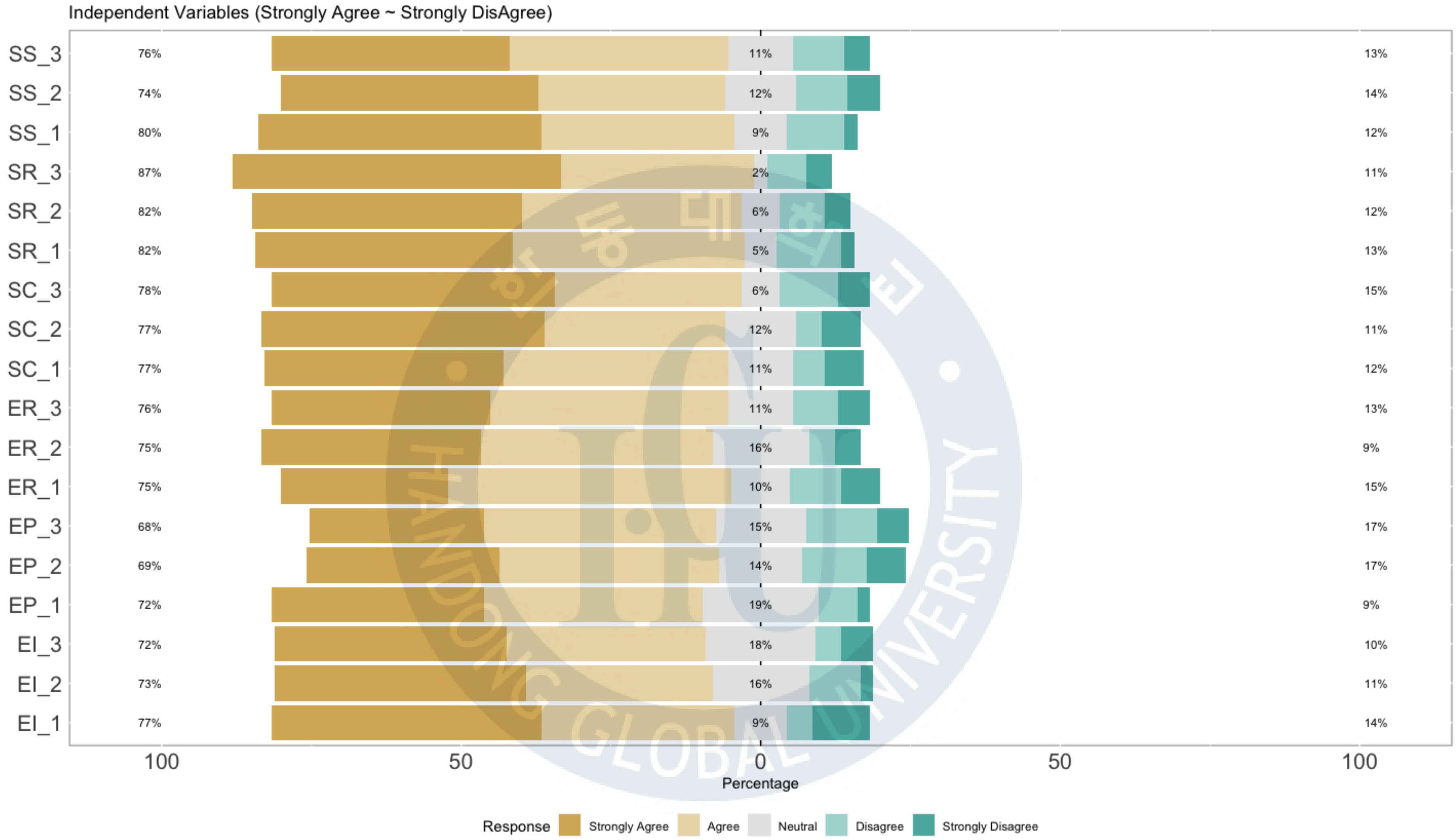

Figure 4-1 Distribution of Independent Variables

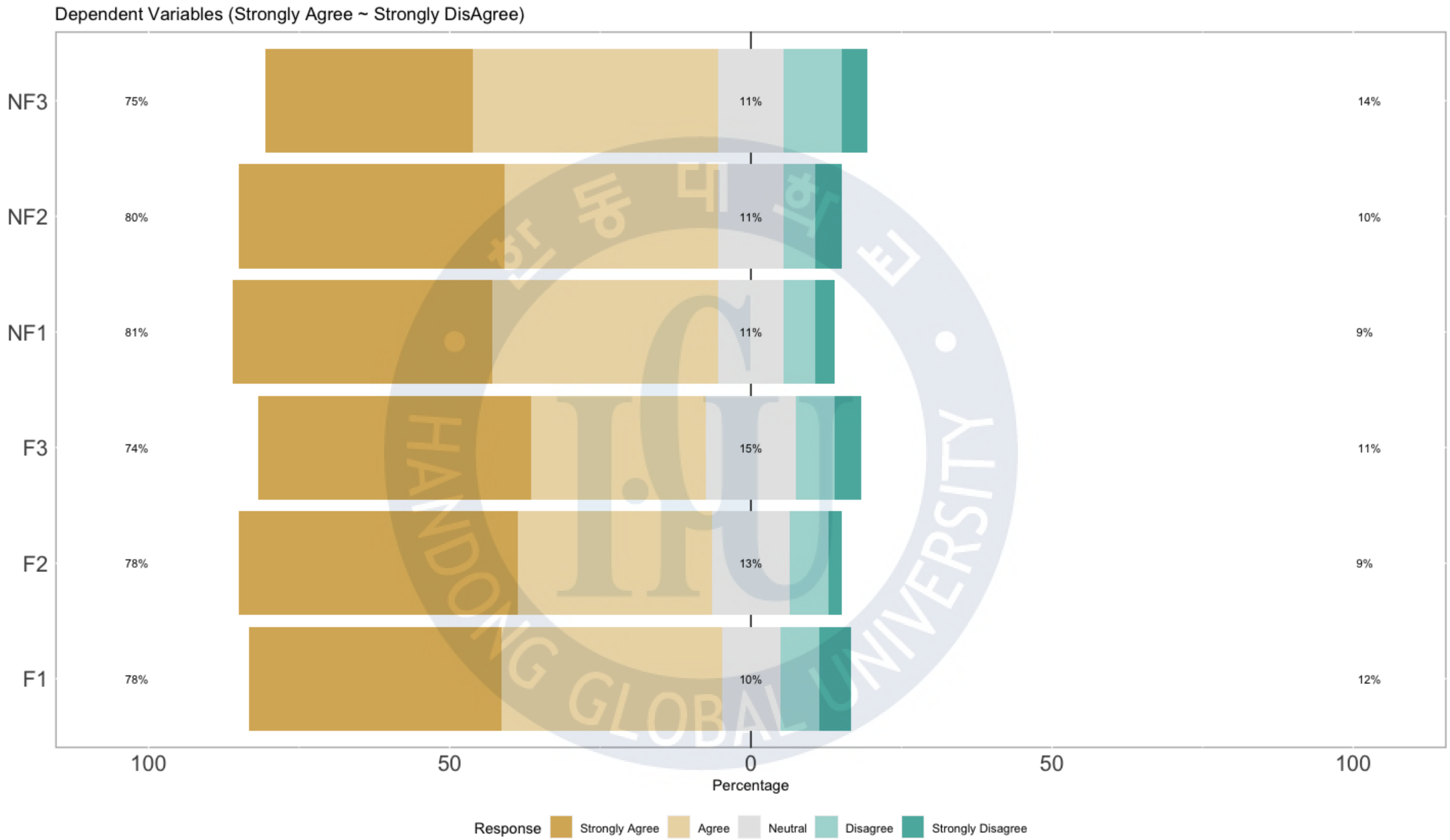

Figure 4-2 Distribution of Dependent Variables

Grounded in the distribution of these measurement items, Table 4-4 shows a descriptive analysis of the research sample that presents the mean, standard deviation, the degree of skewness, and kurtosis.

Table 4-4 Descriptive Analysis of Research Sample (N=93)

Main Factors	Sub Factors	Mean	Standard Deviation	Skewness	Kurtosis
Entrepreneurial Orientation	Innovativeness	2.01	0.91	1.18	0.65
	Proactiveness	2.17	0.94	0.73	-0.28
	Risk-Taking	2.17	0.94	0.73	-0.28
Social Capital	Structural Dimension	1.97	0.97	1.35	1.02
	Cognitive Dimension	1.96	1.06	1.31	0.79
	Relational Dimension	1.84	0.98	1.64	1.95
Startup Performance	Financial Performance	1.93	0.99	1.25	0.81
	Non-Financial Performance	1.96	0.93	1.24	1.13

Table 4-4 shows the average, standard deviation, skewness, and kurtosis in each factor. In entrepreneurial orientation, the mean of innovativeness is the lowest point with 2.01 (SD = 0.91), proactiveness and risk-taking have the same result of the mean and standard deviation with 2.17 (SD = 0.94). In Social Capital, the mean of the relational dimension is the lowest point with 1.84 (SD = 0.98). The cognitive dimension displays the highest point of standard deviation with 1.06 (Mean = 1.96). The mean of the structural dimension is 1.97 (SD = 0.97). In startup performance, the mean of financial performance is 1.93 (SD = 0.99) and 1.96 (SD = 0.93) the non-financial performance is.

When the skewness is greater than 2 in absolute value, the variable can be considered as asymmetrical about its mean. When the kurtosis is greater than or equal to 3, then the distribution of the variable is different from a normal distribution that means the tendency to produce outliers (Westfall & Henning, 2013). In skewness, the lowest point is 0.73 and the highest point is 1.64. In kurtosis, the lowest point is 0.28 and the highest point is 1.95. To sum up, the skewness and kurtosis of innovativeness in entrepreneurial orientation are the highest value, the relational dimension in social capital is the highest value, and non-financial performance in startup performance is the highest value.

In order to figure out the differences of variables by the characteristics derived from sampling, age, gender, position, education, firm age, firm size, startup experience, venture capital experience, and industry experience are used to describe the surveyed data.

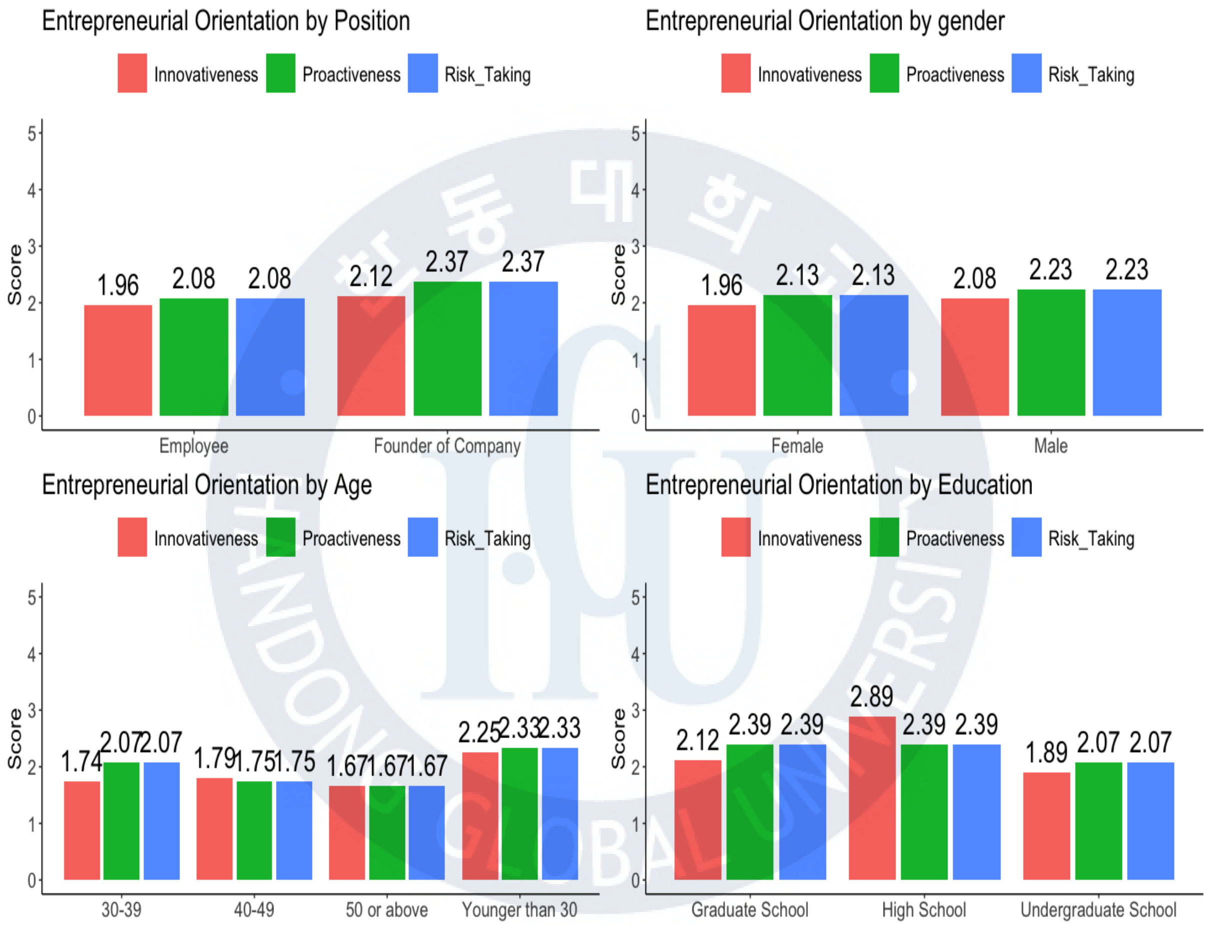

Figure 4-3 Differentiation of Entrepreneurial Orientation by Demographics

Figure 4-3 shows the differentiation of entrepreneurial orientation by demographics. The smaller score means the strong tendency of each factor. In position, an employee (N=63) has more entrepreneurial orientation than the employer (N=30). In gender, female (N=56) is stronger than male (N = 37). In comparison between 30-39 (N = 34) and younger 30 (N = 49), the tendency of 30-39 is much stronger than the group of younger than 30. In education, the people from undergraduate school

(N = 64) have the strongest tendency of entrepreneurial orientation than both from graduate school (N = 23) and high school (N = 6).

The differentiation of entrepreneurial orientation by control variables is also visualized in Figure 4-4. In the firm age, the group of 3-4 years (N = 27) have better scores than others. In the firm size, the group of 5-9 members (N = 16) is the highest score in innovativeness, the group of 10-14 (N = 16) members is better than other groups in terms of proactiveness and risk-taking. When researching startup experience, the people (N = 62) who have no experience have the more tendency of entrepreneurial orientation than the startup experienced people (N = 31). In venture capital experience, the respondents (N = 19) experienced in venture capital is higher than the respondents (N = 74). Last, in industrial experience, the respondents (N = 44) who had worked in the same business field with the current company had a stronger tendency of entrepreneurial orientation than the other group (N = 49)

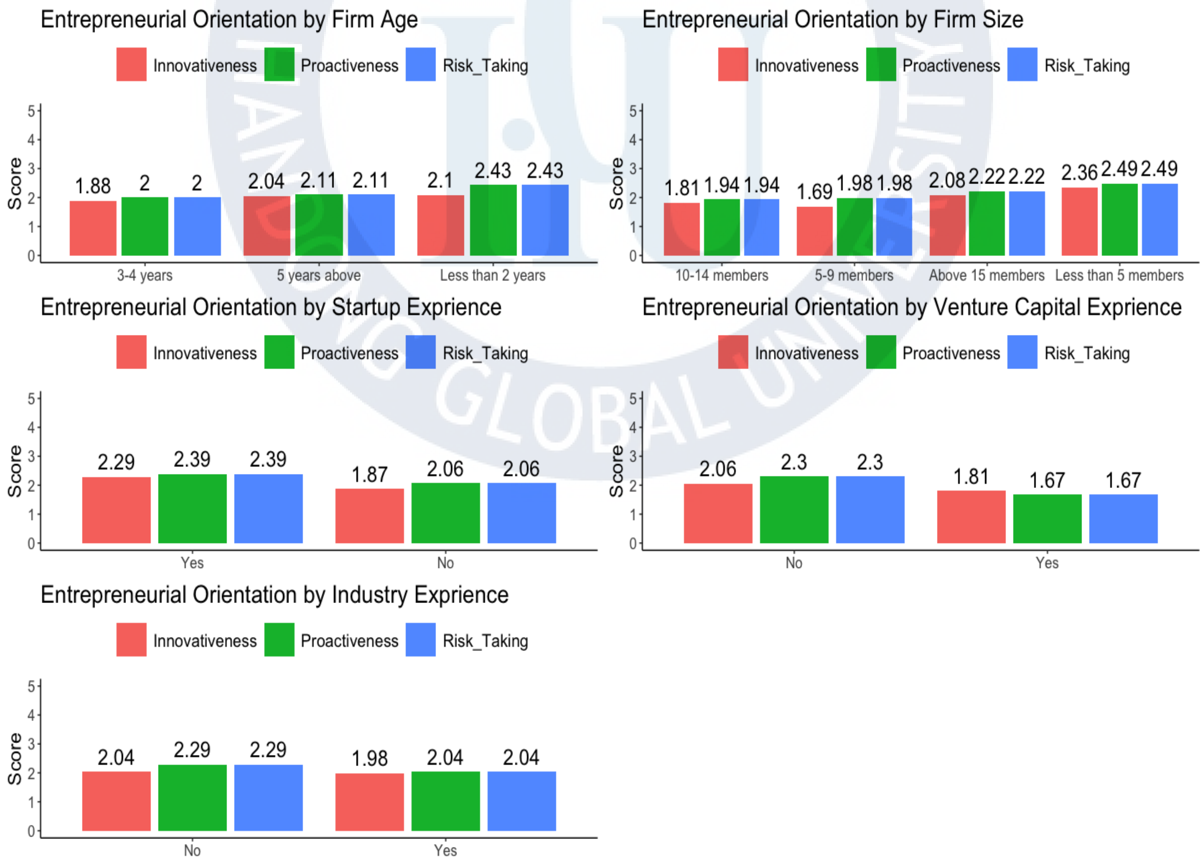

Figure 4-4 Differentiation of Entrepreneurial Orientation by Control Variables

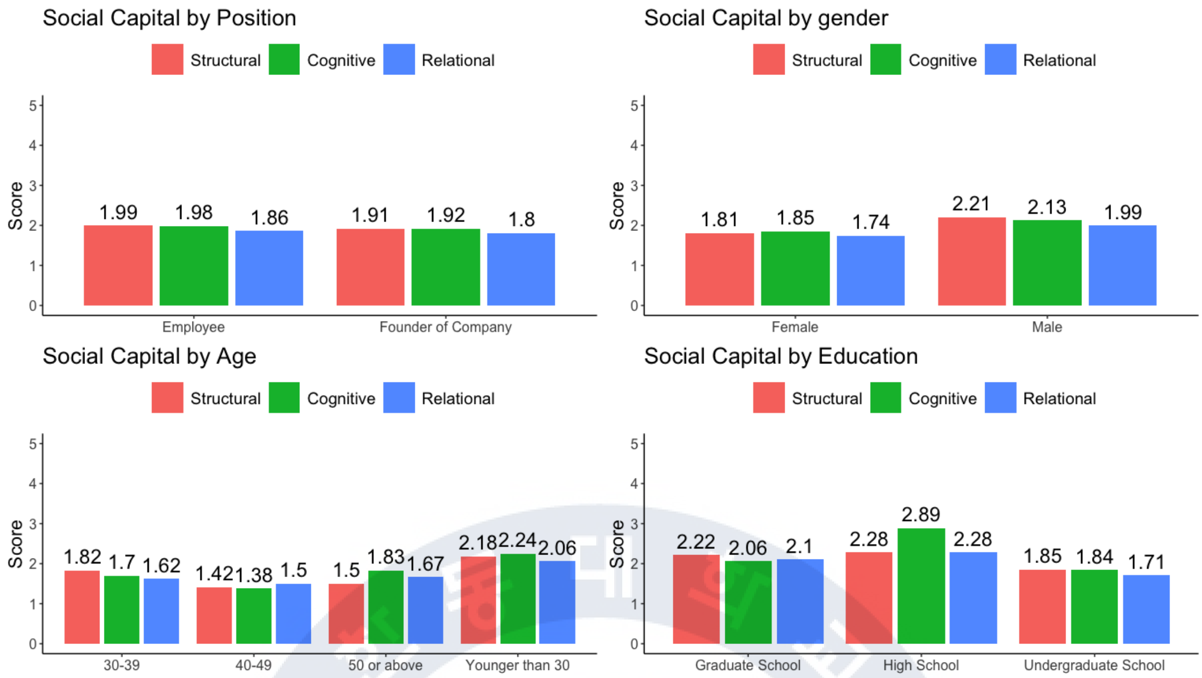

Figure 4-5 Differentiation of social capital by demographics

Figure 4-5 shows the differentiation of social capital by demographics. In position, the founder of the company (N=30) has a stronger social capital than the employee (N=63). This result is the opposite of the result of entrepreneurial orientation. In gender, female (N=56) is stronger than male (N = 37). In comparison between 30-39 (N = 34) and younger 30 (N = 49), the social capital tendency of 30-39 is much stronger than the group of younger than 30. However, In education, the people from undergraduate school (N = 64) have the strongest tendency of social capital than both from graduate school (N = 23) and high school (N = 6).

Figure 4-6 shows the differentiation of social capital by control variables. In the firm age, the group of 3-4 years (N = 27) have better scores than others. In the firm size, the group of 5-9 members (N = 16) is generally the highest score in social. In startup experience, the score is interestingly very similar to whether respondents experienced in a startup company or not. In the cognitive dimension, however, respondents without startup experience (N = 62) have a stronger tendency of social capital than the group with the experience (N = 31). In both venture capital experience and industry experience, the experienced respondents are more likely to have stronger social capital than the other group.

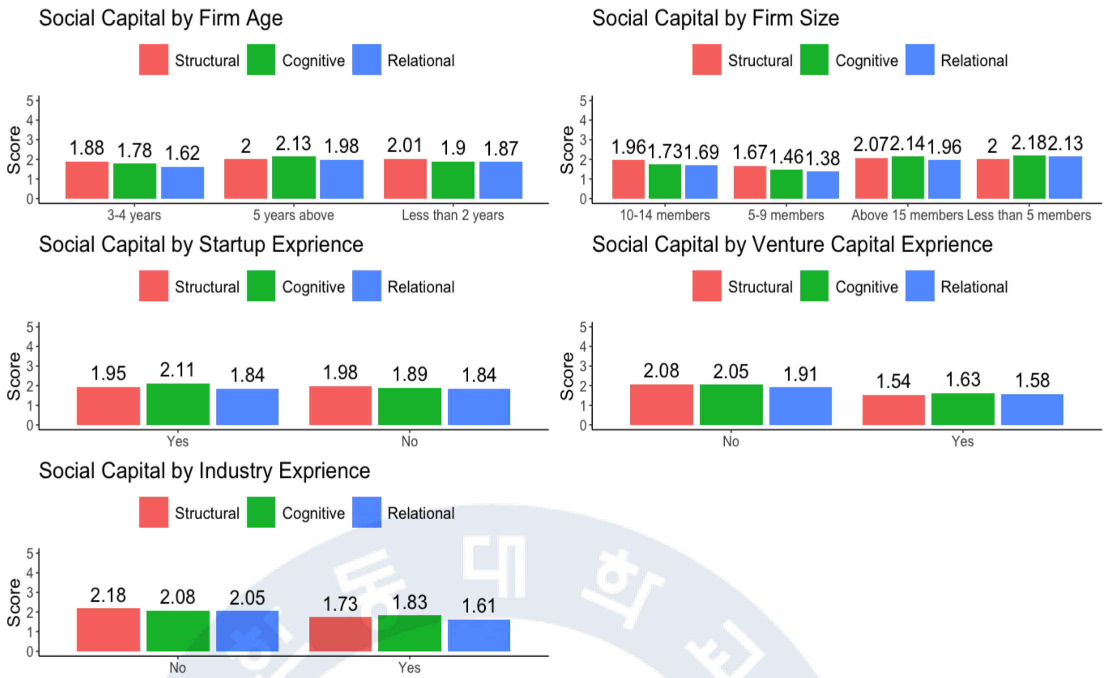

Figure 4-6 Differentiation of Social Capital by Control Variables

Figure 4-7 shows the differentiation of startup performance by demographics. In position, the score in employees (N=63) is higher than the founder of the company (N=30). In gender, the score in females (N=56) is higher than males (N = 37). In comparison between 30-39 (N = 34) and younger 30 (N = 49), the score in 30-39 is higher than the group of younger than 30. In education, the score in undergraduate school (N = 64) is higher than both from graduate school (N = 23) and high school (N = 6).

Figure 4-8 shows the differentiation of startup performance by control variables. In the firm age, the score in 3-4 years (N = 27) is higher than the others. However, in firm size, the group of 10-14 members (N = 16) is the highest score in social. In startup experience, the respondents (N = 62) who had no experience in startup points higher than other groups. In venture capital experience, the score from the respondents (N = 19) who experienced venture capital is higher than the other group. In industry experience, the score with respondents (N = 44) who worked in a similar business field is higher than in other groups.

Since others in the business field are uncertain and the highest rank, it is inappropriate to conduct the descriptive analysis. Accordingly, the descriptive analysis of this area was excluded.

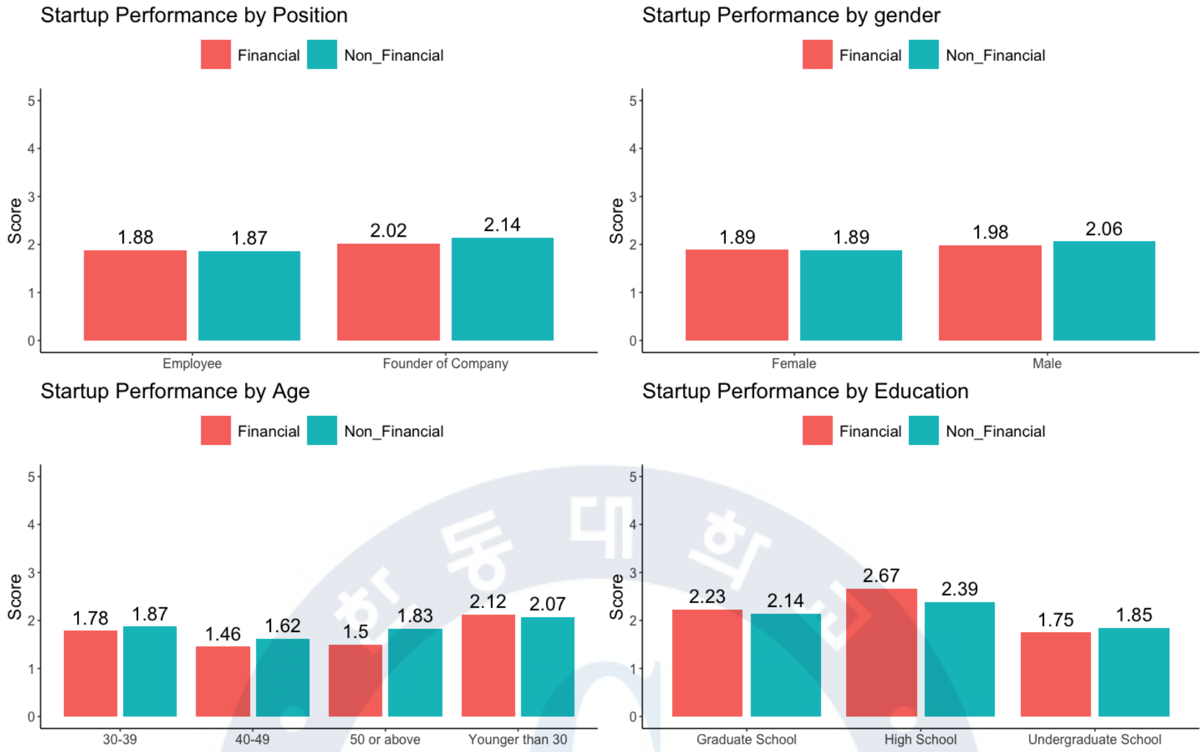

Figure 4-7 Differentiation of Startup Performance by demographics

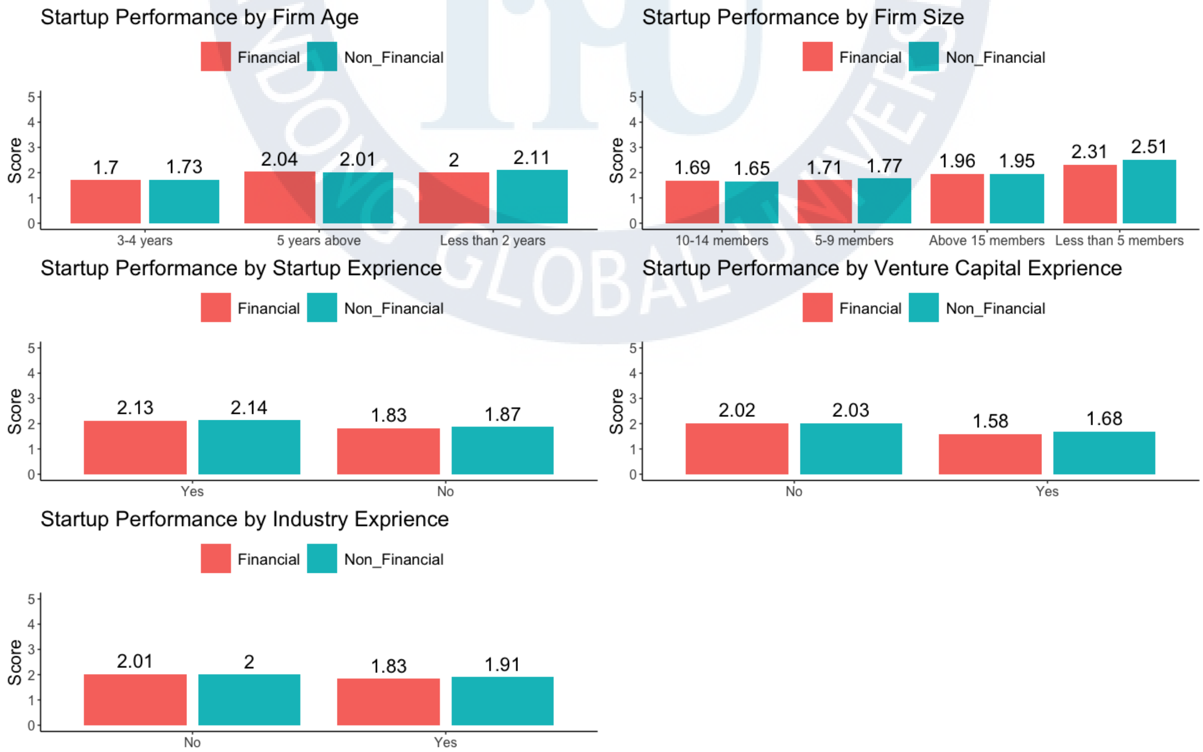

Figure 4-8 Differentiation of Startup Performance by Control Variables

4.3. Analysis of Reliability and Validity

In the course of analysis utilizing PLS-SEM, it is important to test whether the indicator variables are reliable and valid for each variable if the variables are constructed directly by the researcher. The objective of testing reliability and validity is to reduce the measurement error as much as possible (Hair et al., 2016). It is that without reliability and validity, the estimation of the final model using PLS-SEM has no meaning since the measurement error consistently occurs. Accordingly, the reliability and validity of the scales must be ensured through the evaluation process for the measurement item in advance to perform PLS-SEM.

4.3.1. Internal Consistency

Reliability refers to the degree of consistency in the measurement of latent variables. The purpose of reliability analysis is how consistent it is to measure latent variables. In other words, the same results can be obtained by repeating the measures intended to be. This is called internal consistency reliability (Shin 2018). Internal consistency reliability consists of three elements – Cronbach Alpha, rho_$_A$(P_A), Composite Reliability. The criterion is as followed in Table 4-5.

Table 4-5 Acceptance Criteria of Internal Consistency

Criterion	Acceptance Criteria	Reference
Cronbach Alpha	Below 0.60 Low reliability 0.60-0.70 Acceptable reliability 0.70-0.80 Desirable reliability 0.80-0.90 High reliability	Cronbach(1951), Nunnally and Bernstein (1994)
Dijkstra-Henseler's rho_$_A$(P_A)	P_A > 0.7: Acceptable reliability	Dijkstra and Henseler(2015)
Composite Reliability	Below 0.60 A lack of reliability 0.60 – 0.70 Acceptable in exploratory re-search 0.70 – 0.90 Acceptable reliability Above 0.95 Not desirable	Werts et al. (1974), Nunnally and Bernstein (1994)

(Source: Shin, 2018, p. 195)

Cronbach's α test was conducted to ensure reliability. As described in Figure 4.9, all the factors are above the 0.7 level which means desirable reliability. The highest factor was entrepreneurial orientation whereas the lowest factor was structural dimension and structural dimension. The details are followed in Table 4-6.

Table 4-6 Results of Cronbach α

Latent Variable		Items	Cronbach's $\boldsymbol{\alpha}$
Entrepreneurial Orientation		9	0.900
Social Capital	Structural Dimension	3	0.838
	Cognitive Dimension	3	0.889
	Relational Dimension	3	0.892
Startup Performance	Financial Performance	3	0.889
	Non-Financial Performance	3	0.838
	Total	24	

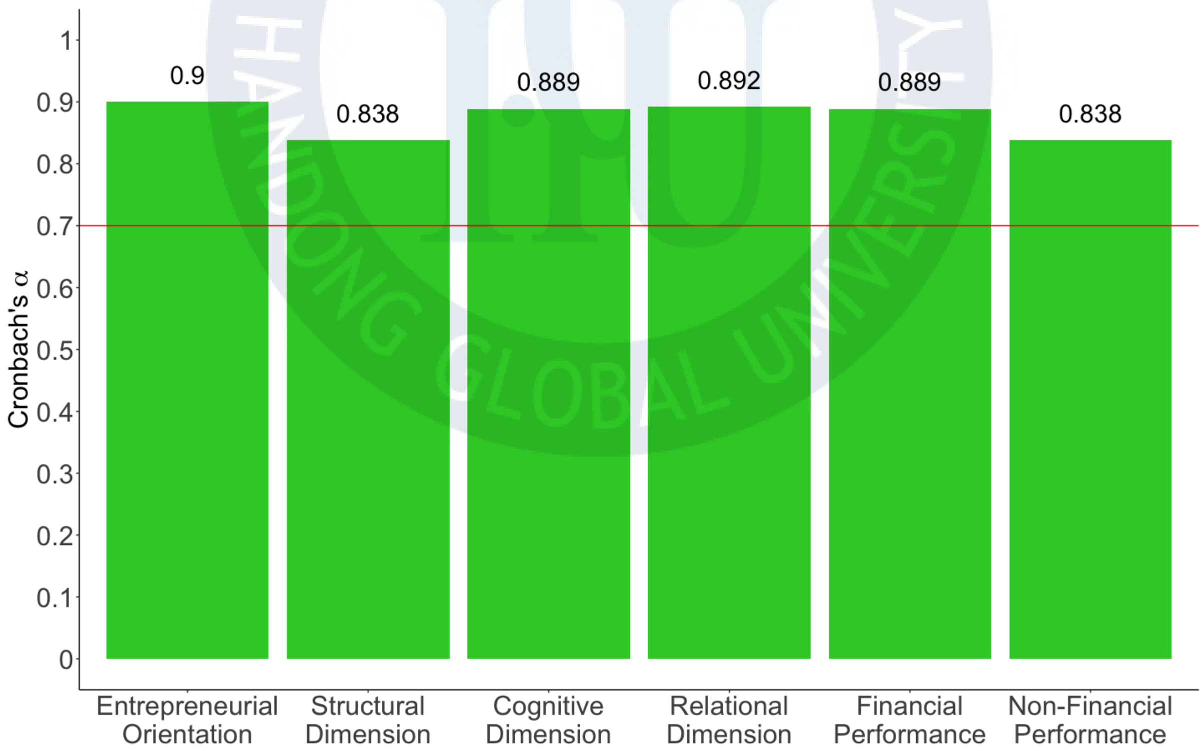

Figure 4-9 Cronbach's α

The result of the D-H rho_A(P_A) test shows its reliability with the following result. As described in Figure 4-10, all the factors are above the 0.7 level which means desirable reliability. The highest factor was entrepreneurial orientation, where the lowest factor was the structural dimension. The details are followed in Table 4-7.

Table 4-7 Results of D-H rho_A(PA)

Latent Variable		Items	D-H rho_A(P_A)
Entrepreneurial Orientation		9	0.910
Social Capital	Structural Dimension	3	0.839
	Cognitive Dimension	3	0.891
	Relational Dimension	3	0.893
Startup Performance	Financial Performance	3	0.892
	Non-Financial Performance	3	0.864
	Total	24	

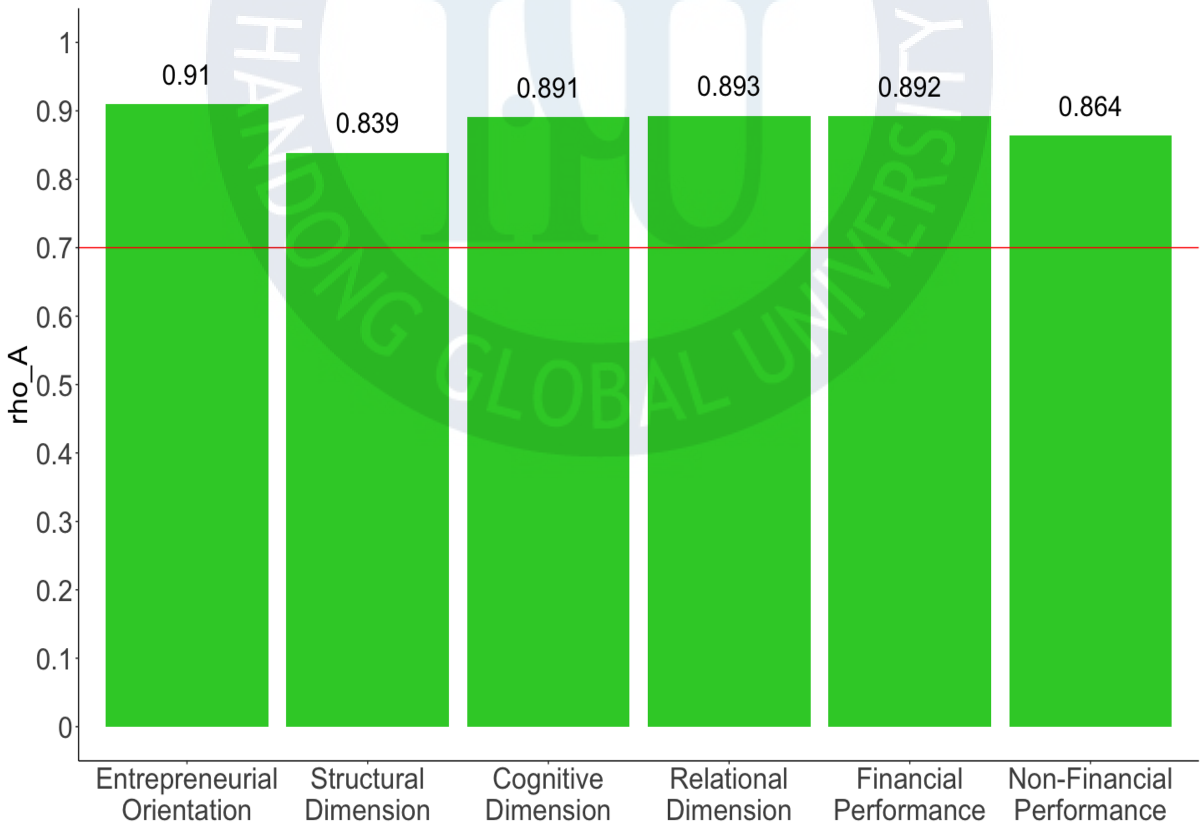

Figure 4-10 D-H rho_A(PA)

The result of Composite Reliability (P_C) shows its reliability with the following result. As described in Figure 4-10, all the factors are above the 0.7 level which means desirable reliability. The highest factor was the relational dimension, whereas the lowest factor was the structural dimension. The details are followed in Table 4-8.

Table 4-8 Result of Composite Reliability (P_C)

Latent Variable		Items	P_C
Entrepreneurial Orientation		9	0.919
Social Capital	Structural Dimension	3	0.902
	Cognitive Dimension	3	0.931
	Relational Dimension	3	0.933
Startup Performance	Financial Performance	3	0.931
	Non-Financial Performance	3	0.902
	Total	24	

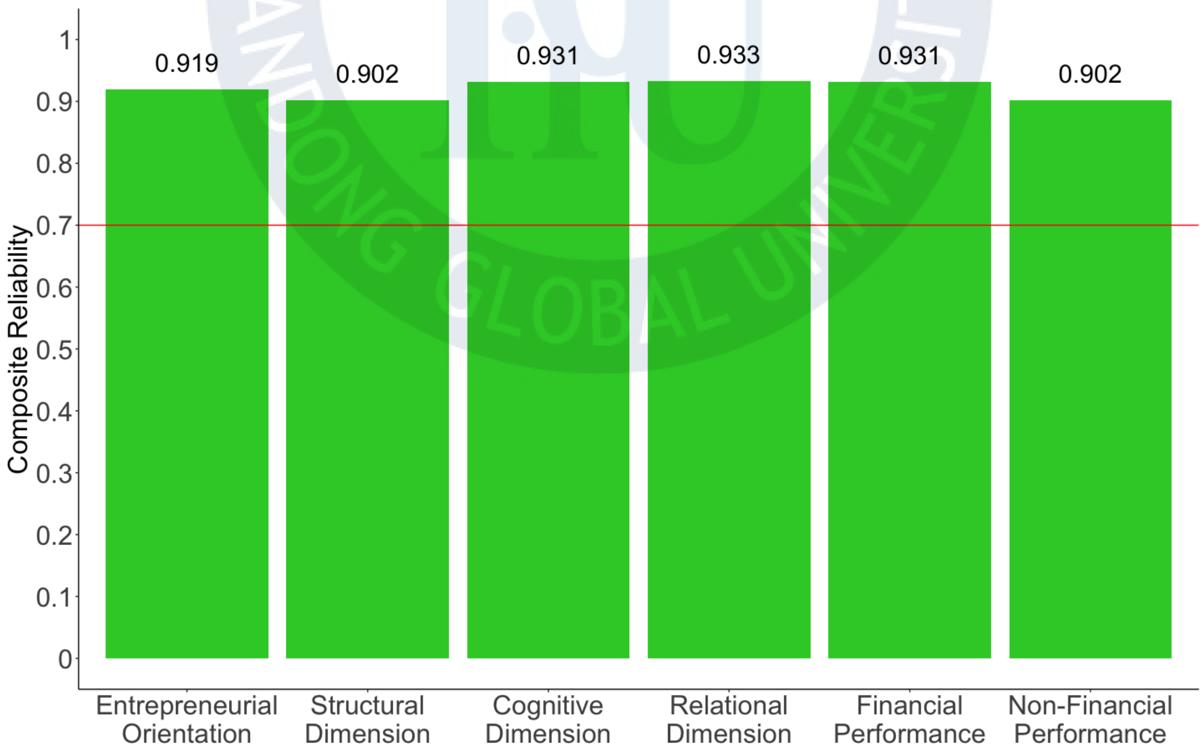

Figure 4- 11 Composite Reliability (PC)

4.3.2. Convergent Validity

Convergent Validity means the extent to which a measure positively correlates with alternative measures of the same construct. To evaluate the convergent validity of constructs, researchers have used the outer loadings of the indicators and the average variance extracted (AVE). The outer loading relevance testing should be considered with the statistical significance and the size of the outer loading, which is commonly called indicator reliability (Hair, et al., 2016). The acceptance criteria of convergent validity are described in Table 4-9.

Table 4-9 Acceptance Criteria of Convergent Validity

Criterion	Acceptance Criteria	Reference
Outer Loading Relevance ($\mathcal{L}$)	See Figure 4-12	Bagozzi, Yi and Philipps (1991), Hair et al., (2016)
Indicator Reliability	Above 0.5 Desirable reliability	Chin (1998)
AVE (Average Variance Extracted)		Fornell and Larcker (1981)

(Source: Shin, 2018, p. 195)

The value 0.5 in indicator reliability and AVE means that 50% of the variance of an indicator can be explained by the latent variable. On outer loading relevance, researchers often get weaker outer loadings (< 0.70) in social science studies. It recommends eliminating carefully indicators according to figure 4-12 (Hair et al., 2016).

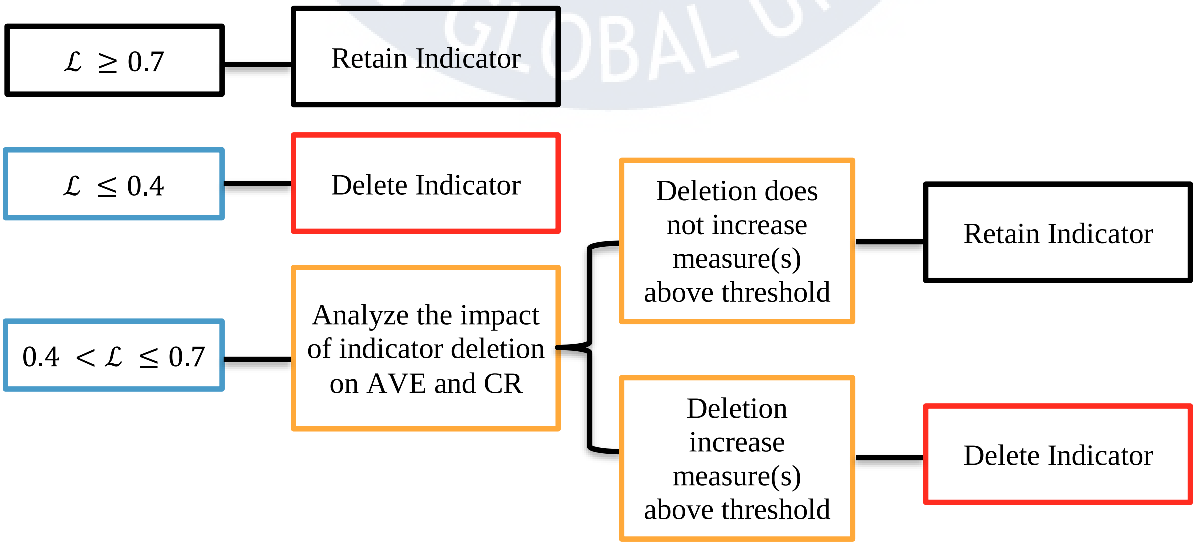

Figure 4-12 Outer Loading Relevance Testing

The first output of convergent validity is as followed in Table 4-10.

Table 4-10 Results of Convergent Validity Using SmartPLS

Latent Variable	Indicator	Outer Loadings	Indicator Reliability	AVE
Entrepreneurial Orientation	EI_1	0.793	0.629	0.559
	EI_2	0.650	0.423	
	EI_3	0.673	0.453	
	EP_1	0.714	0.510	
	EP_2	0.687	0.472	
	EP_3	0.707	0.500	
	ER_1	0.846	0.716	
	ER_2	0.821	0.674	
	ER_3	0.810	0.656	
Structural Dimension	SS_1	0.894	0.865	0.819
	SS_2	0.841	0.738	
	SS_3	0.872	0.856	
Cognitive Dimension	SC_1	0.883	0.817	0.756
	SC_2	0.911	0.819	
	SC_3	0.920	0.630	
Relational Dimension	SR_1	0.905	0.780	0.819
	SR_2	0.913	0.830	
	SR_3	0.902	0.846	
Financial Performance	F1	0.93	0.819	0.822
	F2	0.859	0.834	
	F3	0.925	0.814	
Non-Financial Performance	NF1	0.904	0.799	0.756
	NF2	0.905	0.707	
	NF3	0.794	0.760	

(Source: SmartPLS Algorithm)

As shown in the table, the value of EI_2, EI_3, and EP_2 is lower than 0.7, thus these indicators are analyzed if deleting them increases above threshold (> 0.7). It turns out that AVE and CR are still above the threshold, these indicators are deleted and re-test it again. After the second trial, EP_3 results in 0.693, so repeat it again until all of the outer loadings above the threshold. The final indicators and results on entrepreneurial orientation are described in Table 4-11 and Figure 4-13.

Table 4-11 Final Indicators for Convergent Validity

Latent Variable	Indicator	Outer Loadings	Indicator Reliability	AVE
Entrepreneurial Orientation	EI_1	0.834	0.696	0.756
	ER_1	0.902	0.814	
	ER_2	0.861	0.741	
	ER_3	0.879	0.773	

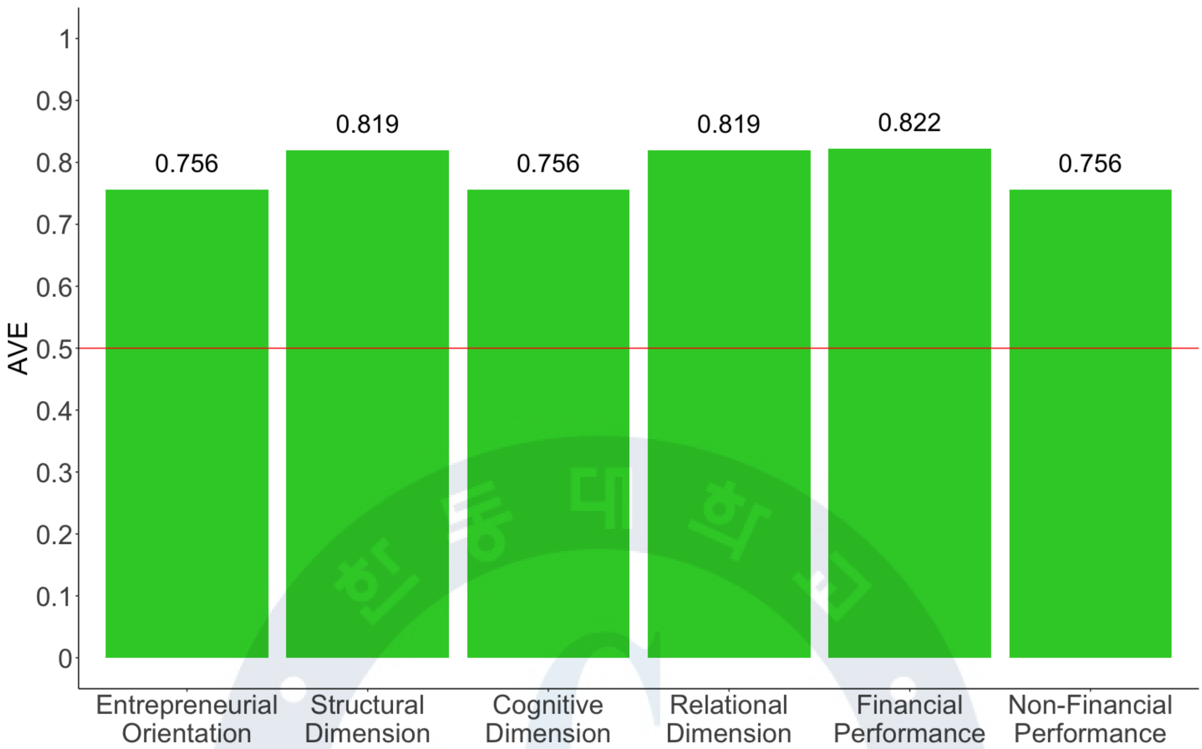

Figure 4-13 The final Result of Average Variance Extracted (AVE)

4.3.3. Discriminant Validity

Discriminant validity means the degree to which the latent variables can be well distinguished, which means that the latent variables are independent of each other. It should be little or no correlation between the results when the different latent variables are measured by the measurement method (Shin, 2018). To evaluate discriminant validity, researchers have relied on two measures – Fornell-Larcker Criterion (Fornell and Larcker, 1981) and the cross-loading (Chin, 1998). The acceptance criteria of discriminant validity are as followed in Table 4-12.

Table 4-12 Acceptance Criteria of Discriminant Validity

Criterion	Acceptance Criteria	Reference
Fornell-Larcker Criterion	Acceptable: The AVE square root of each latent variable is greater than the highest correlation between the latent variables	Fornell and Larcker (1981)
Cross loadings	Acceptable: The outer loadings of an indicator on a construct should exceed all its cross-loadings with other constructs.	Chin (1998)

(Source: Shin, 2018, p. 196)

The first criterion is assessed by the SmartPLS algorithm function. In this research, SC_2, SR_3 and F3 indicators are removed in order to meet Fornell-Larcker Criterion. In this research, all square roots of AVE exceeded the off-diagonal elements in their corresponding row and column in Table 4-13. The bolded values in Table 4-13 mean the square roots of the AVE and the non-bolded values means the inter-correlation value between constructs. All off-diagonal elements are lower than square roots of AVE confirming that Fornell and Larker's criterion is met.

Table 4-13 Fornell-Larcker

	SC	EO	FP	NFP	SR	SD
Cognitive Dimension	**0.926**					
Entrepreneurial Orientation	0.856	**0.869**				
Financial Performance	0.809	0.774	**0.916**			
Non-Financial Performance	0.782	0.826	0.866	**0.869**		
Relational Dimension	0.848	0.839	0.823	0.840	**0.931**	
Structural Dimension	0.778	0.763	0.669	0.659	0.801	**0.869**

The cross-loadings, which is the second assessment for discriminant validity, involves examining items and comparing them to all construct correlations. The loading indicators on the assigned construct should be higher than all loading on other constructs. The output of cross-loadings produced by the SmartPLS algorithm function is presented in Table 4-14. As you can see, all of the indicators used in this research loaded higher against their intended latent variable compared to other variables.

Table 4-14 Cross Loadings

	SC	EO	FP	NFP	SR	SS
EI_1	0.763	**0.834**	0.641	0.685	0.73	0.594
ER_1	0.761	**0.903**	0.696	0.764	0.755	0.664
ER_2	0.724	**0.860**	0.699	0.727	0.737	0.719
ER_3	0.728	**0.879**	0.655	0.693	0.694	0.674
F1	0.824	0.773	**0.930**	0.814	0.781	0.636
F2	0.645	0.636	**0.901**	0.77	0.722	0.587
NF1	0.697	0.758	0.808	**0.906**	0.813	0.633
NF2	0.784	0.819	0.818	**0.908**	0.794	0.641
NF3	0.522	0.532	0.599	**0.787**	0.541	0.403
SC_1	**0.919**	0.769	0.703	0.684	0.739	0.742
SC_3	**0.933**	0.814	0.792	0.762	0.828	0.701
SR_1	0.766	0.775	0.748	0.776	**0.929**	0.772
SR_2	0.812	0.786	0.784	0.789	**0.933**	0.72
SS_1	0.676	0.652	0.586	0.564	0.706	**0.896**
SS_2	0.680	0.704	0.547	0.516	0.692	**0.841**
SS_3	0.672	0.635	0.61	0.634	0.691	**0.870**

Overall, the reliability and validity tests conducted on the measurement model suggested the final indicators which are valid to fit to be used as estimate parameters in the structural model. The final results of reliability and validity from the selected indicators are described in Table 4-15.

Table 4-15 Final Results of Reliability and Validity

Latent Variable	Indicators	Convergent Validity			Internal Consistency		
		Outer loadings	Indicator Reliability	AVE	Cronbach α	rho_A(P_A)	CR
Entrepreneurial Orientation	EI_1	0.834	0.696	0.756	0.892	0.893	0.925
	ER_1	0.902	0.814				
	ER_2	0.860	0.740				
	ER_3	0.880	0.774				
Structural Dimension	SS_1	0.894	0.799	0.755	0.838	0.838	0.903
	SS_2	0.845	0.714				
	SS_3	0.867	0.752				
Cognitive Dimension	SC_1	0.903	0.815	0.858	0.835	0.840	0.923
	SC_3	0.914	0.835				
Relational Dimension	SR_1	0.902	0.814	0.867	0.846	0.846	0.929
	SR_2	0.930	0.865				
Financial Performance	F1	0.894	0.799	0.838	0.809	0.824	0.912
	F2	0.845	0.714				
Non-Financial Performance	NF1	0.867	0.752	0.755	0.838	0.872	0.902
	NF2	0.915	0.837				
	NF3	0.925	0.856				

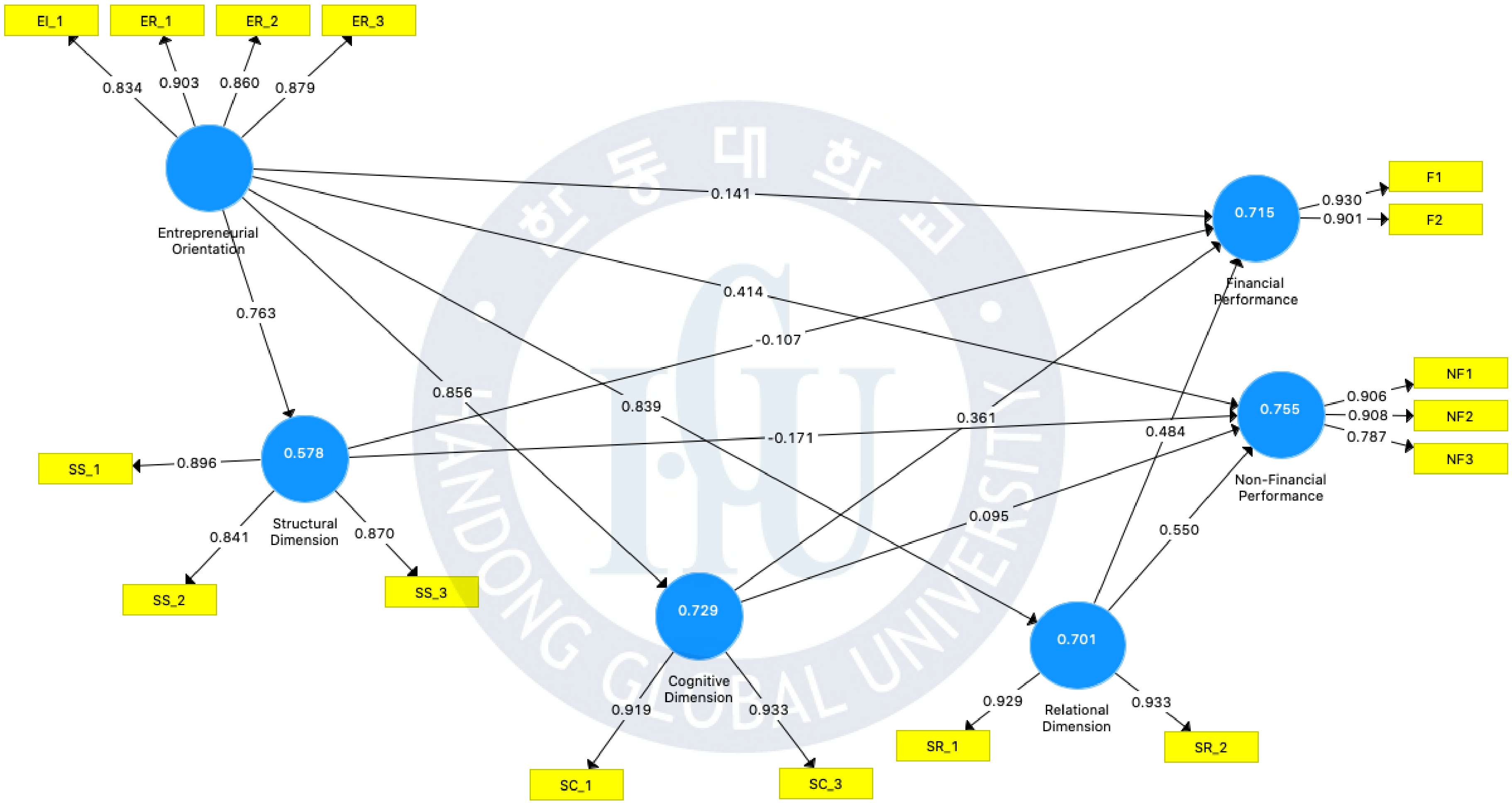

Figure 4-14 Results of PLS-SEM Algorithm

4.4. Analysis of PLS-SEM Structural Model Results

The first step is to examine the structural model for multicollinearity. The structural model is grounded in OLS regressions of each endogenous latent variable on its corresponding predecessor constructs as similar to the multiple regression (Hair, et al., 2016). After multicollinearity, the structural model in PLS-SEM needs to measure as followed in Table 4-16.

Table 4-16 Acceptance Criteria of PLS-SEM Structural Model

Steps	Criteria	Content	Acceptance Level	Reference
1	Multicollinearity	Collinearity between each set of predictor variables.	VIF $<$ 5: Collinearity VIF $\geq$: No Collinearity	-
2	Coefficient of Determination	A measure of in-sample predictive power	R^2 0.25: weak R^2 0.50: moderate R^2 0.75: substantial	Rigdon, 2012, Sarstedt et al., 2013
3	f^2 effect size	The substantive impact of the specified exogenous construct on the R^2 value of endogenous constructs	$\leq$ 0.02: No effect 0.02: small 0.15: medium 0.35: large effects	Cohen, 1988
4	Q^2 Predictive Relevance	Indicator of the model's out-of-sample predictive power.	Q^2 values $>$ 0 - model has predictive relevance Q^2 values $\leq$ 0 - model indicates lack of predictive relevance	Geisser, 1974, Stone, 1974
5	Significance of the path coefficients	The relevance of significant relationships which represent the hypothesized relationships among the constructs.	Using the bootstrap confidence interval Critical values - t value 1.65: $\alpha = 0.1$ - t value 1.96: $\alpha = 0.05$ - t value 2.57: $\alpha = 0.01$	-

(Source: Shin, 2018, p. 248)

4.4.1. Multicollinearity

SmartPLS algorithm generates T-statistics for significant tests of the structural model using a procedure called bootstrapping. A large number of subsamples (e.g., 5000) in this procedure are taken from the original sample with a replacement to give bootstrap standard errors. Multicollinearity is tested by using Variance Inflation Factors (VIF). As shown in Table 4-17, since all values are less than 5.0, it can conclude that there is no collinearity in this model.

Table 4-17 VIF values on Inner Model

	SC	EO	FP	NFP	SR	SS
Cognitive Dimension			4.957	4.957		
Entrepreneurial Orientation	1		4.605	4.605	1	1
Relational Dimension			4.88	4.88		
Structural Dimension			3.161	3.161		

4.4.2. Coefficient of Determination (R^2)

The coefficient of determination (R^2) is commonly used to measure the structure model and calculated as the squared correlation between an endogenous construct's actual and predicted values. The coefficient indicates the amount of variance in the endogenous constructs explained by all of the exogenous constructs linked to it. As seen in Table 4-16, a larger R^2 value increases the predictive ability of the structural model. In this dissertation, the SmartPLS algorithm function generates the R^2 values as followed in Table 4-18. The adjusted R^2 of the endogenous construct, NFP (0.755) indicates the substantial predictive power, the SC (0.729), FP (0.715), SR (0.701), and SS (0.578) represent the moderate predictive power. The R^2 criterion is met, and the structural model can be an adequate predictive ability.

Table 4-18 Results of R^2

Endogenous Constructs	R^2	R^2Adjusted	T Statistics	P Values
Cognitive Dimension (SC)	0.732	0.729	12.447	0.000
Financial Performance (FP)	0.728	0.715	12.069	0.000
Non-Financial Performance (NFP)	0.765	0.755	14.476	0.000
Relational Dimension (SR)	0.704	0.701	11.872	0.000
Structural Dimension (SS)	0.582	0.578	5.256	0.000

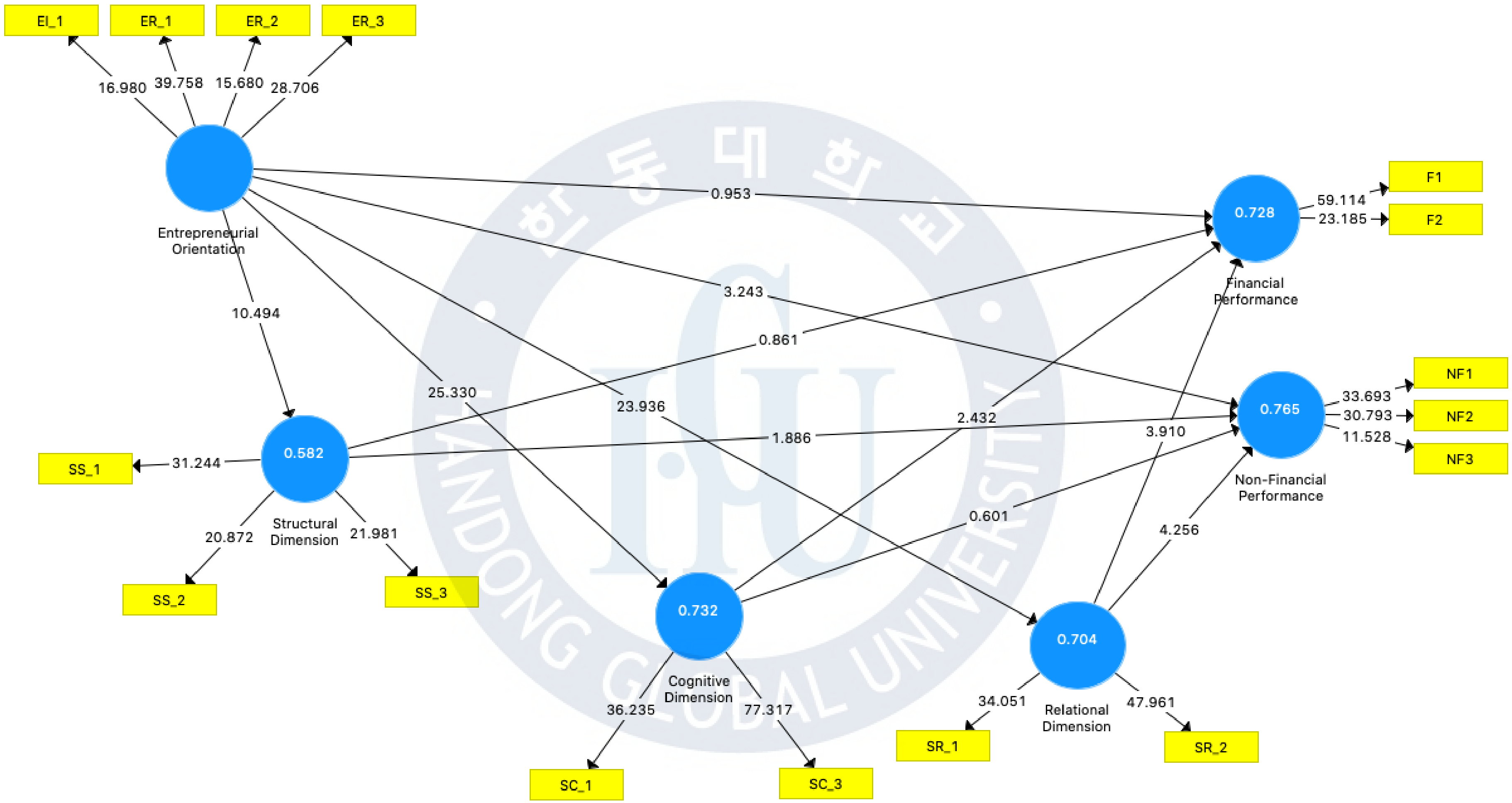

Figure 4-15 Results of Bootstrapping

4.4.3. Effect Size (f^2)

The effect size (f^2) in the evaluation of the structural model is a measure of the extent to which the R^2 of the endogenous construct contributes to the specified exogenous construct. The journal editors and reviewers are increasingly encouraged to refer to this measure. The following Table 4-19 shows the results of the effect size (f^2). The relational dimension is the highest contribution to the endogenous constructs, both FP (0.176) and NFP (0.264) in this research. The cognitive dimension contributes to FP (0.096) which is between small and medium in financial performance while there is no effect on non-financial performance (0.008). Entrepreneurial orientation indicates that there is no effect on financial performance (0.016), whereas it is medium to non-financial performance (0.158). Likewise, the structural dimension has no effect on financial performance (0.013) and respectively small effects on non-financial performance (0.039).

Table 4-19 Results of f^2

Endogenous Constructs	SC	EO	FP	NFP	SR	SS
Cognitive Dimension			0.096	0.008		
Entrepreneurial Orientation	2.732		0.016	0.158	2.377	1.394
Financial Performance						
Non-Financial Performance						
Relational Dimension			0.176	0.264		
Structural Dimension			0.013	0.039		

4.4.4. Blindfolding and Predictive Relevance (Q^2)

In addition to R^2 values as a criterion of predictive accuracy, Q^2 is also necessary to be examined by researchers (Geisser, 1974; Stone, 1974). The value of Q^2 can be obtained by using the blindfolding procedure of omission distance D in the SmartPLS algorithm. If the value of Q^2 is larger than zero, then it means that the endogenous latent variable has the predictive relevance of the structural model for a particular dependent construct. The results of Q^2 is as followed in Table 4-20. As shown in Table 4-20, all of the values are larger than zero, thus it can support the predictive relevance of the current structural model.

Table 4-20 Results of Q^2

Endogenous Constructs	SSO	SSE	Q^2 (=1-SSE/SSO)
Cognitive Dimension	186	71.459	0.616
Entrepreneurial Orientation	372	372	
Financial Performance	186	76.412	0.589
Non-Financial Performance	279	125.62	0.55
Relational Dimension	186	73.475	0.605
Structural Dimension	279	157.855	0.434

4.4.5. Path Coefficients and Hypothesis Testing

After running the PLS-SEM algorithm, the path coefficients are obtained for the structural model relationship. Since the coefficients have standardized values approximately between -1 and +1, the estimated path coefficients close to +1 represent strong positive relationships, whereas the negative values mean negative relationships. If the value is close to zero, then it is not significantly different from zero (Hair, et al., 2016).

To conduct a significance test, t-value, p-value, and confidence interval are commonly used. When t value is larger than the critical value, it is that the coefficient is statistically significant. In social science studies, the critical value 1.96 (significance level = 5%) is commonly used for two-tailed tests. Likewise, p value is mostly used to assess significance levels. When assuming a significance level of 5%, the p value must be smaller than 0.05. The bootstrap confidence interval can test whether a path coefficient is significantly different from zero. If a confidence interval for an estimated path coefficient does not include zero, the hypothesis that the path equals zero is rejected, and we assume a significant effect. In this research, all of the suggested ways are provided to test the significance of all structural model relationships. The final results of hypothesis testing are shown in Table 4-21.

Table 4-21 Summary of Hypothesis Testing

No	Hypothesis Path	Path Coefficients	T Statistics	P Values	95% CI		Result
H1-1	EO -> FP	0.141	0.94	0.347	-0.153	0.436	No
H1-2	EO -> NFP	0.414	3.28	0.001	0.161	0.656	Yes
H2-1	EO -> SS	0.763	10.191	0.000	0.579	0.875	Yes
H2-2	EO -> SC	0.856	25.231	0.000	0.762	0.905	Yes
H2-3	EO -> SR	0.839	23.769	0.000	0.747	0.890	Yes
H3-1	EO -> SS -> FP	-0.082	0.862	0.389	-0.284	0.102	No
H3-2	EO -> SC -> FP	0.309	2.45	0.014	0.063	0.557	Yes
H3-3	EO -> SR -> FP	0.406	3.774	0.000	0.192	0.612	Yes
H3-4	EO -> SS -> NFP	-0.131	1.811	0.070	-0.302	-0.011	No
H3-5	EO -> SC -> NFP	0.081	0.604	0.546	-0.189	0.336	No
H3-6	EO -> SR -> NFP	0.461	4.141	0.000	0.252	0.689	Yes
CV	Firm Age -> FP	0.032	0.516	0.606	-0.088	0.156	No
	Firm Age -> NFP	0.063	1.169	0.242	-0.046	0.164	No
	Firm Size -> FP	0.020	0.369	0.712	-0.090	0.122	No
	Firm Size -> FP	0.076	1.581	0.114	-0.026	0.163	No
	WE -> FP	-0.062	0.819	0.413	-0.212	0.090	No
	WE -> NFP	-0.035	0.557	0.557	-0.161	0.087	No

EO : Entrepreneurial Orientation
SC : Structural Dimension
CC : Cognitive Dimension
RC : Relatinoal Dimension
FP : Financial Performance
CV: Control Variables
NFP : Non-Financial Performance
WE : Work Experience

To test the proposed hypotheses and the structural model in this research, the path coefficients (β) between latent variables are evaluated. A path coefficient value should be at least 0.1 to account for a certain impact within the model (Hair et al., 2016). As shown in Table 4-21, seven of the proposed hypotheses are supported. H1-1, H3-1, H3-4, and H3-5 are not significantly supported in this research. Supported hypotheses are significant at the level of 0.05 and possess a path coefficient value (β) ranging from 0.141 to 0.839.

In this research, entrepreneurial orientation is positively relates to non-financial performance (β = 0.414; $p <$ .001), supporting H1-2. Entrepreneurial orientation is positively related to the structural dimension of social capital (β = 0.763; $p <$ .000), supporting H2-1. Entrepreneurial orientation is positively related to the cognitive dimension of social capital (β = 0.856; $p < .000$), supporting H2-2. Entrepreneurial orientation is positively relates to relational dimension of social capital (β = 0.839; $p < .000$), supporting H2-3. On the basis of these results, it would conclude that entrepreneurial orientation is positively related to social capital in general.

The cognitive dimension in social capital mediates in the relationship between entrepreneurial orientation and financial performance (β = 0.309; $p < .014$), supporting H3-2. The relational dimension in social capital mediates in the relationship between entrepreneurial orientation and financial performance (β = 0.406; $p < .000$), supporting H3-3. The relational dimension in social capital mediates in the relationship between entrepreneurial orientation and non-financial performance (β = 0.461; $p < .000$), supporting H3-6. To summarise, the relational dimension in social capital is the main key construct to mediate the relationship between entrepreneurial orientation and startup performance. Meanwhile, the age of startups, the size of startups, and the relationship between whether or not startup experience has been found to be unverified in this research.

4.4.6 Total Effects

Researchers need to evaluate not only one construct's direct effect on another but also its indirect effects via one or more mediating constructs. The total effect is the sum of direct and indirect effects (Hair et al., 2016). This is significantly meaningful in relevant studies focusing on exploring the differential impact of several independent constructs on dependent constructs via one or more mediating variables. The comparison result between the path coefficient and total effects is described in Table 4-22.

Table 4-22 Total Effect

Hypothesis Path	Type	Original Sample (O)	Sample Mean (M)	Standard Deviation (STDEV)	T Statistics (O/STDEV)	P Values
EO -> FP	Direct Effect	0.141	0.136	0.15	0.94	0.347
EO -> FP	Total effects	0.774	0.771	0.063	12.341	0.000
EO -> NFP	Direct Effect	0.414	0.412	0.126	3.28	0.001
EO -> NFP	Total Effects	0.826	0.823	0.043	19.216	0.000

As shown in Table 4-22, there is an indirect effect between two constructs between entrepreneurial orientation and startup performance. With the path coefficient (β = 0.309; $p < .014$), the cognitive dimension is mediated in the relationship between entrepreneurial orientation and financial performance. This finding can be interpreted in which entrepreneurial orientation affects financial performance when visible vision and goal are shared by staff within startups. Accordingly, the hypothesis 1-1 can be supported in terms of indirect-only mediation in which the indirect effect is significant but not the direct effect (Hair et al., 2016).

4.4.7 Discussion of Research Results

Social capital becomes an important business strategy for firms. This research is in line with previous work on the role of social capital, such as Adler and Kwon (2002), and Nahapiet and Ghoshal (1998) by investigating the relationship among entrepreneurial orientation, social capital, and startup performance. This research proves that social capital can be a business strategy that leads to greater efficiency at startups located in developing countries.

First, entrepreneurial orientation has a significantly positive influence on financial performance. This finding is consistent with previous studies by (Hongyun et al., 2019), who analyzing the small-medium enterprises operating in Ghana as a developing country that entrepreneurial orientation has a positive relationship with firm performance and (Key and Rahman, 2018) who was examining spin-off and symbiosis company in Malaysia that entrepreneurial orientation is statistically related to startup success.

Second, entrepreneurial orientation has a significantly positive influence on all of the dimensions of social capital. This finding is consistent with previous studies conducted by (Lee et al., 2019) who was analyzing the performance of startups in Korea, empirically proving that entrepreneurial orientation had a positive effect on the structural, cognitive, and relational dimensions of social capital.

Third, social capital mediates significantly but partially mediates the relationship between entrepreneurial orientation and startup performance. As shown in Table 4-21, the relational dimension of social capital was the only key construct that mediates the relationship between entrepreneurial orientation and startup performance. This finding is in contrast to other studies conducted in Korea (Lee, 2019). The proposed hypothesis of the relational dimension is insignificant in Korea which is differentiating from the Philippines.

To sum up, entrepreneurial orientation positively affects both social capital and startup performance in terms of total effects in the Philippines like other developing countries. However, it becomes arguable which dimension of social capital mediates in the relationship between entrepreneurial orientation and startup performance.

Chapter 5. Conclusions

The main objective of this research is to explore the status quo of startups in the Philippines, to verify the existing research of startup performance with entrepreneurial orientation and social capital in the country, and to develop a predictive model which dimension of social capital is to mediate the relationship between entrepreneurial orientation and startup performance. Based on surveying startups (N = 93) in the Philippines, this research conducts the mediating effects of social capital, the relationship between entrepreneurial orientation and startup performance and reviews the direct effect of entrepreneurial orientation on social capital and startup performance.

Startup performance as the dependent variable was used and divided into two aspects – perceived financial and non-financial performance surveyed from each respondent. The entrepreneurial orientation of startups was divided into one single factor including innovativeness, proactiveness, and risk-taking. Social capital consists of structural, cognitive, and relational dimensions to verify that they had a positive impact on startup performance. It also validated how social capital had a mediating effect on the relationship between entrepreneurial orientation and startup performance.

The research models and hypotheses for measurement and analysis were designed and samples were selected. To describe the samples, R (version 4.0.2) was used to calculate and visualize using tidyverse (version 1.3.0) package them for descriptive analysis. Verification of hypotheses was conducted using SmartPLS (version 3.2.3), the applied methodologies were internal consistency reliability, convergent validity, discriminant validity, collinearity assessment, coefficient of determination, effect size, predictive relevance, and structural model path coefficients generated by SmartPLS.

To summarise all these findings after testing hypotheses, entrepreneurial orientation has a positive impact on social capital and startup performance. In mediating effects, social capital is partially but positively mediates in the relationship between entrepreneurial orientation and startup performance. The following chapters will be explained in detail with implications and its limitations.

5.1. Practical Implications

The research results have the following theoretical importance. First, it indicated that entrepreneurial orientation contributes to the development of social capital among startups in the Philippines that represent similar results with other researches (Wiklund & Shepherd, 2003; Zhou et al., 2005; Yoon, 2014; Kee & Rahman, 2018; Lee et al., 2019; Hongyun et al., 2019). This result is linked with the research (Marino et al., 2002) which was analyzing the relationship between entrepreneurial orientation and strategic alliance formation, suggested that the firms located in Finland, Greece, Indonesia, Mexico, Netherlands, and Sweden with stronger entrepreneurial orientation will use strategic alliances, which can be interpreted as social capital in some manner, more extensively. In other words, startups with higher entrepreneurial orientation enhance enterprising decision making and innovative behavior, developing and diversify networks that can integrate and allocate resources, and accumulating trust and norms within networks.

Second, it empirically proved again that although it failed to prove the direct effect of entrepreneurial orientation on financial performance, it still worked when it comes to total effects of entrepreneurial orientation on both financial and non-financial performance. This finding is important for startups and the government in the Philippines. Several studies conducted in the Philippines were related to small-medium companies either operating over old companies or locally based companies (Ashill & Chadee, 2015; Reyes, 2019). Accordingly, this finding is meaningful in terms of trying to conduct research, targeting startups related to IT-based companies in the Philippines. It proved again that entrepreneurial orientation still works operating the companies beyond SMEs in the Philippines.

Third, it suggested a research perspective that comprehensively deals with entrepreneurial orientation, sub-dimensions of social capital, and startup performance and confirmed the partially mediated effects of the relational dimension of social capital. Through this research, this finding is also important for new foreign investors and partners. The main key feature when it comes to startup performance, the range of networks of founder and employee is more important than their vision and communication skill for startup performance which is different from the startups in Korea.

5.2. Limitation of Research and Future Research Direction

With the above significance, this study has the following limitations. First, there is a limit to the survey. Contrary to the original intention of the research, it was somewhat difficult to objectify it theoretically because data were collected only in the form of randomization online without conducting on the field due to external environmental issues related to Covid-19 around the world. In future studies, it is necessary to collect data offline so that data can be obtained systematically.

Second, as similar to the first limitation, it did not consider the demographics of respondents. Since many IT companies such as e-commerce and fintech are operating in the Philippines, it needs to divide startups into IT startups and non-IT startups. Corresponding to the type of startups, social capital might be differently mediated in entrepreneurial orientation and firm performance. Also, it needs to adapt the theory of strategic orientation, which consists of entrepreneurial orientation, market orientation, and technological orientation, so that future research will figure out which orientation is more significant for the startups in the Philippines.

Third, it was positive to try measuring financial and non-financial performance in this research. However, this is less objective because it was measured on the basis of the perceived and subjective evaluation of each respondent. Therefore, future research needs to develop an objective evaluation index that can enhance the generalization of startup performance.

Fourth, it suggests focusing on the structural dimension of social capital rather than other dimensions. It is difficult to figure out whether strong ties or weak ties mediate in the relationship between entrepreneurial orientation and startup performance. It is unfolded that when I interviewed with an employer who runs e-commerce for three years, he mostly shared that strong connections such as family and friends are more helpful than the others at the first stage. However, when business runs well, the importance of weak connections become increased to expand (FabStorePH, 2020).

In the future, research on mathematical-statistical analysis using such surveys will require continuous supplementation, development, and effort of standardized measurement tools and methods suitable for the situation in the Philippines.

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APPENDIX

Kamusta,

I, Evan Jung, am a student majoring in Global Development & Entrepreneurship at graduate school in Handong Global University.

As part of my research thesis titled, "A Study on the Interplay of Entrepreneurial Orientation and Social Capital in Shaping Startup Performance: The cases of the Philippines", this questionnaires are constructed.

I will appreciate if you could complete the following table. Any information obtained in connection with this study that can be identified with you will remain confidential.

This survey will take about 5 minutes.

Thank you for your cooperation in the survey and we look forward to your company's prosperity and growth. Maraming Salamat po.

June, 2020

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Entrepreneurial Orientation

Strongly Agree (1)	Agree (2)	Neutral (3)	Disagree (4)	Strongly Disagree (5)

Innovativeness

Item	Measurement item	(1)	(2)	(3)	(4)	(5)
E1	*In general, my collogues including the founder of my firm generally favor,* A Strong emphasis on R&D, technological leadership, and innovations					
E2	*How many new lines of products or services has your firm marketed in the past 1-3 years?,* Very many new lines of products or services					
E3	Changes in product or service lines have usually been quite dramatic in the past 1-3 years					

Proactiveness

Item	Measurement item	(1)	(2)	(3)	(4)	(5)
E4	*In dealing with its competitors, my firm…,* Typically initiates actions which competitors then respond to our company.					
E5	*In dealing with its competitors, my firm…,* My firm is often the first business to Very many new lines of products or services.					
E6	*In dealing with its competitors, my firm…,* Changes in product or service lines have usually been quite dramatic					

Risk-Taking

Item	Measurement item	(1)	(2)	(3)	(4)	(5)
E7	*In general, the founder and managers of my firm have…* A strong proclivity for high-risk projects (with chances of very high returns)					
E8	*In general, the founder and managers of my firm believe* Owing to the nature of the environment, bold, wide-ranging acts are necessary to achieve the firm's objectives.					
E9	*When confronted with decision-making situations involving uncertainty, my firm Typically,* adopts a bold, aggressive posture in order to maximize the probability of exploiting potential opportunities.					

Social Capital

Strongly Agree (1)	**Agree (2)**	**Neutral (3)**	**Disagree (4)**	**Strongly Disagree (5)**

Structural Dimension

Item	Measurement item	(1)	(2)	(3)	(4)	(5)
S1	*In Business/Non-Business,* My team members maintain close social relationships with other members					
S2	*In Business/Non-Business,* I know some members in the team on a personal level.					
S3	*In Office,* In general, members would be happy to spend the rest of their career with collogues					

Cognitive Dimension

Item	Measurement item	(1)	(2)	(3)	(4)	(5)
C1	*In my firm,* My team shares the same ambitions and vision with other units at work					
C2	*In my firm,* People in our unit are enthusiastic about pursuing the collective goals and missions of the whole organization.					
C3	*In Office,* My team get most of our valuable and shared information including technical know-how related to supplying our product/service from other team members.					

Relational Dimension

Item	Measurement item	(1)	(2)	(3)	(4)	(5)
R1	*In this relationship,* My team members will always keep the promises they make to one another.					
R2	*In this relationship,* My team members are truthful in dealing with one another.					
R3	*When confronted with solving any kind of problems,* I believe that members in my team would help me if I need					

Startup Performance

Strongly Agree (1)	Agree (2)	Neutral (3)	Disagree (4)	Strongly Disagree (5)

The perceived Financial Performance

Item	Measurement item	(1)	(2)	(3)	(4)	(5)
F1	*In the recent 1-3 years,* My team is perceived that my firm's market share is increasing.					
F2	*In the recent 1-3 years,* My team is perceived that my firm's cash flow is significantly in change					
F3	*In the recent 1-3 years,* My team is perceived that my firm's sales are increasing					

The perceived Non-Financial Performance

Item	Measurement item	(1)	(2)	(3)	(4)	(5)
NF1	*In the recent 1-3 years,* We believe that our company has achieved its current goal or vision to some extent					
NF2	*In the recent 1-3 years,* Other competitors and venture capitalist objectively values our company's growth potential.					
NF3	*In the recent 1-3 years,* My firm's patent has been innovated and registered in relevant organization.					

Control Variables

1. How many years has it been since your company started? (Firm Age)

① Less than 2 years ② 3-4 years ③ 5 years above

2. How many staffs, full-time, are currently working in your company? (Firm Size)

① Less than 5 members ② 5-9 members ③ 10-14 members ④ Above 15 members

3. Have you ever had a start-up experience before? (WE1)

① No, I don’t have

② Yes, I have startup experience, one time

③ Yes, I have startup experience, two times

④ Yes, I have startup experience more then three times

4. Have you ever worked in the venture capital firm? (WE2)

① Yes ② No

5. Have you ever worked in the same industry or related to industry? (WE3)

① Yes ② No

Demographic Survey

1. Please write down your company's name

2. Please write down your company's website

3. Please indicate your gender

① Female ② Male

4. Please indicate your job position

① Founder of Company ② Employee (Director or Manager)

5. Please indicate the category that includes your age

① Younger than 30 ② 30-39 ③ 40-49 ④ 50 or above

6. What is the highest degree or level of school you have completed? If currently enrolled, highest degree received.

① High School ② Undergraduate School ③ Graduate School

7. In which sectors are you working in?

Real Estate and Household		E-Commerce	
Media and Entertainment		Travel	
Online to Offline Commerce		Others	
Transportation/Automotive		Fintech	
Enterprise Services		Energy	
Hardware		Education	
Logistics		Medical and Healthcare	